Neurocritical Care

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Learning Objectives

1. Identify pertinent pathophysiologic and laboratory changes that acutely occur after neurologic injuries and require therapeutic intervention.
2. Describe monitoring devices commonly used in neurocritical care patients that help develop and optimize treatment strategies.
3. Develop an evidence-based treatment strategy for neurocritical care patients that optimizes patient outcomes and reduces the risk of adverse drug effects and drug interactions.
4. Recommend a monitoring plan to assess response to therapeutic regimens and specific therapeutic goals for neurocritical care patients.
5. Develop new plans of care for neurocritical care patients according to therapeutic and adverse outcomes and progress toward therapeutic goals.

Abbreviations in This Chapter

ADH	Antidiuretic hormone
CNS	Central nervous system
CPP	Cerebral perfusion pressure
CSF	Cerebrospinal fluid
CSWS	Cerebral salt-wasting syndrome
EEG	Electroencephalogram
GCS	Glasgow Coma Scale
ICH	Intracerebral hemorrhage
ICP	Intracranial pressure
ICU	Intensive care unit
MAP	Mean arterial pressure
MRSE	Methicillin-resistant *Staphylococcus epidermidis*
PSH	Paroxysmal sympathetic hyperactivity
RSE	Refractory status epilepticus
SAH	Subarachnoid hemorrhage
SCI	Spinal cord injury
SIADH	Syndrome of inappropriate antidiuretic hormone
SDOH	Social determinants of health
TBI	Traumatic brain injury
VTE	Venous thromboembolism

Self-Assessment Questions

Answers and explanations to these questions may be found at the end of this chapter.

1. A 56-year-old woman is hospital day 4 after her acute aneurysmal subarachnoid hemorrhage (SAH). She is oriented and following commands. Laboratory values reveal a serum sodium of 128 mmol/L. Other serum chemistry values include potassium (K) 3.9 mEq/L, chloride 103 mEq/L, bicarbonate 27 mEq/L, blood urea nitrogen (BUN) 10 mg/dL, and serum creatinine (SCr) 1.0 mg/dL. Her urinary output is 1–2 mL/kg/hour, and her fluid balance has been +435 mL during the past 24 hours (currently receiving 0.9% sodium chloride at 125 mL/hour). Which is the best initial therapy for this patient's hyponatremia?
   A. Tolvaptan 20 mg orally daily.
   B. 1.5% sodium chloride infusion at 125 mL/hour.
   C. Water restriction to less than 1.5 L/day.
   D. No treatment indicated right now.

2. A 27-year-old woman is admitted with an acute ventriculoperitoneal shunt failure and associated infection. She has no significant medical history and no allergies to medications. A lumbar puncture reveals cerebrospinal fluid (CSF) white blood cell count (WBC) 34×10^3 cells/mm^3, red blood cell count (RBC) 1×10^3 cells/mm^3, protein 78 mg/dL, and glucose 21 mg/dL. The cultures grow methicillin-resistant *Staphylococcus epidermidis* (MRSE). Her shunt is externalized. Despite 4 days of intravenous vancomycin (most recent vancomycin trough was 17.7 mcg/mL), the CSF continues to grow MRSE. Which is the most appropriate intraventricular antimicrobial regimen to initiate for this patient's refractory ventriculitis?
   A. Give vancomycin 10 mg intraventricularly daily.
   B. Give gentamicin 5 mg intraventricularly daily.
   C. Give ampicillin 50 mg intraventricularly daily.
   D. No antimicrobials should be given intraventricularly for this patient.

3. A 25-year-old man is admitted after a two-story fall from a ladder. The initial computed tomography (CT) scan of his brain reveals a large right temporal subdural hematoma, an overlying skull fracture, and a left temporal contusion. His post-resuscitation Glasgow Coma Scale (GCS) score is E1-M4-V1T. An intracranial pressure (ICP) monitor is placed with an opening pressure of 32 mm Hg and a cerebral perfusion pressure (CPP) of 53 mm Hg. Serum laboratory values include sodium (Na) 141 mEq/L, K 3.6 mEq/L, BUN 8 mg/dL, SCr 1.1 mg/dL, glucose 178 mg/dL, WBC 14.8 × 10^3 cells/mm^3, pH 7.46, and partial pressure of carbon dioxide (P_{CO_2}) 34 mm Hg. Which supportive care issue is most relevant to the appropriate treatment of a patient with a severe traumatic brain injury (TBI)?
   A. Avoid enteral nutrition for the first 5–7 days because of the lack of gastrointestinal (GI) tolerance in severe TBI.
   B. Maintain CPP at 60–70 mm Hg to optimize perfusion and reduce complications.
   C. Provide dextrose 5% or other dextrose-containing intravenous fluids to compensate for the patient's increased metabolic needs.
   D. Initiate high-dose methylprednisolone therapy within 8 hours of injury to reduce cerebral edema.

4. A 69-year-old woman presents to the emergency department with a 30-minute history of difficulty with word finding and left upper-extremity weakness. Her NIH Stroke Scale score is 13. A head CT scan reveals no acute abnormalities. The patient's home medications include lisinopril, carvedilol, warfarin, and atorvastatin. Her medical history includes hypertension, atrial fibrillation, and transient ischemic attacks (diagnosed 6 months ago). Serum laboratory values include Na 145 mEq/L, K 4.0 mEq/L, BUN 18 mg/dL, SCr 1.2 mg/dL, glucose 132 mg/dL, WBC 8.7 × 10^3 cells/mm^3, hematocrit (Hct) 38.9%, platelet count (Plt) 355,000/mm^3, and international normalized ratio (INR) 1.5. Her vital signs include blood pressure 167/98 mm Hg, heart rate 132 beats/minute, oxygen saturation (Sa_{O_2}) 98%, and respiratory rate 14 breaths/minute. Which is the most appropriate next step in this patient's care?
   A. Initiate aspirin 324 mg orally × 1.
   B. Initiate alteplase 0.9 mg/kg intravenously (10% bolus dose, 90% infusion up to 90 mg maximum).
   C. Initiate nicardipine to reduce blood pressure to systolic blood pressure (SBP) less than 140 mm Hg, followed by alteplase 0.9 mg/kg intravenously (10% bolus dose, 90% infusion up to 90 mg maximum).
   D. Initiate vitamin K 10 mg intravenously × 1.

5. An 18-year-old man is admitted to the intensive care unit (ICU) after falling from a tree. Initial trauma screening reveals a C3–C4 fracture and dislocation with an incomplete spinal cord injury (SCI) at the corresponding levels (he has some sensory function bilaterally). The fracture has been reduced, and he arrives in the ICU 6 hours after injury. Which is the most appropriate statement related to initiating high-dose methylprednisolone therapy for this patient's SCI?
   A. May be used because he has an incomplete injury with some sensory function.
   B. Should be used because it will augment spinal perfusion.
   C. Should not be used because of the potential for adverse effects and questionable benefit.
   D. Should not be used because the patient is outside the treatment window.

6. A 27-year-old woman presents with fever, agitation, hypertension, and muscle rigidity. Her drugs-of-abuse screen is negative, and serotonin syndrome is a possible diagnosis. Which home medication is most likely a causative agent for serotonin syndrome?
   A. Buspirone.
   B. Levetiracetam.
   C. Cyproheptadine.
   D. Bupropion.

7. A 58-year-old woman with a Hunt and Hess grade 4 SAH resides in your ICU. She is day 6 after her SAH. Her current medications include 0.9% normal saline at 100 mL/hour, nimodipine 60 mg by feeding tube every 4 hours, norepinephrine 0.05 mcg/kg/minute (5 mcg/minute), famotidine 20 mg intravenously every 12 hours, docusate 250 mg by tube every 12 hours, and morphine as needed for headache. Current laboratory values include Na 144 mEq/L, K 4.1 mEq/L, SCr 0.6 mg/dL, serum osmolality 322 mOsm/L, and Hct 32.3%. Her blood pressure is 167/99 mm Hg, heart rate 133

beats/minute, respiratory rate 18 breaths/minute, Sao_2 99%, and central venous pressure 5 mm Hg. Her most recent transcranial Doppler velocities are mean middle cerebral artery 125/135 (right/left), with a corresponding Lindegaard ratio of 3.5/3.7 (right/left). Her ICP is currently 24 mm Hg. She has a Licox monitor placed in the hemisphere ipsilateral to her aneurysm, which currently reveals a partial pressure of brain tissue oxygen of 14%. She is intubated and on the ventilator. Her GCS score has decreased from 10 to 8 during the past hour. Which treatment would be most appropriate for this patient in the ICU?

A. Verapamil 2.5 mg intra-arterially × 1.
B. 3% sodium chloride 2.5 mL/kg intravenously × 1.
C. 20% mannitol 0.25 g/kg intravenously × 1.
D. 1 unit of packed RBCs.

8. Which sedative in an intubated patient is most desirable for a patient with a diagnosis of SAH currently having vasospasm?

A. Lorazepam 1 mg/hour.
B. Midazolam 4 mg/hour.
C. Morphine 2 mg/hour.
D. Propofol 25 mcg/kg/minute.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
   a. Task A: 1, 3, 6
   b. Task B: 5
2. Domain 2: Therapeutics and Patient Management
   a. Task A: 1–4, 8
   b. Task B: 4, 14
   c. Task C: 2, 3
3. Domain 3: Professional Practice
   a. Task A: 1, 2, 4

I. HYPONATREMIA

A. Epidemiology: Hyponatremia (serum sodium less than 135 mEq/L) is common in patients with a neurologic injury (12%–43%). The most common form of hyponatremia in the neurocritically ill patient is hypotonic hyponatremia (Syndrome of Inappropriate Antidiuretic Hormone [SIADH] and Cerebral Salt-Wasting Syndrome [CSWS]), which is the only type of hyponatremia addressed in this chapter.

B. Diagnosis/Pathophysiology
   1. Laboratory tests (serum sodium) are needed to diagnose hyponatremia.
   2. Urine sodium, urine osmolality, serum osmolality, and measurement of intravascular volume may also help determine the specific pathogenesis for hyponatremia.

C. Causes
   1. Consideration of iatrogenic hyponatremia
   2. Typically caused by an increase in salt-free water or loss of serum sodium

D. Differentiating Between SIADH and CSWS
   1. Typically made by assessing intravascular volume. Patients with SIADH tend to be euvolemic or hypervolemic with hyponatremia because of excessive antidiuretic hormone (ADH) release, whereas patients with CSWS tend to be hypovolemic with hyponatremia because of inappropriate urinary excretion of sodium and extracellular fluid.
   2. Measure intravascular volume using a central venous pressure catheter or similar invasive monitoring.
   3. Noninvasive hemodynamic monitoring devices
   4. Monitor fluid balance, weights, skin turgor
   5. Echocardiogram to estimate ventricular filling pressures

E. Clinical Impact
   1. Hyponatremia may result in increased brain edema and elevated intracranial pressure (ICP).
   2. May cause neurologic symptoms such as delirium, agitation, tremor, seizure, or coma

Table 1. Differential Diagnosis for SIADH and CSWS[a]

	Serum Sodium (mEq/L)	Serum Osmolality (mOsm/L)	Urine Sodium (mEq/L)	Urine Osmolality (mOsm/L)	Intravascular Volume Status
SIADH	< 135	< 285	> 25	> 200	Euvolemia
CSWS	< 135	< 285	> 25	> 200	Hypovolemia

[a]Note: Medications, particularly diuretics, may alter serum or urine measurements of osmolality or sodium concentration.

CSWS = cerebral salt-wasting syndrome; SIADH = syndrome of inappropriate antidiuretic hormone.

Table 2. Hyponatremia Clinical Symptoms by Severity of Hyponatremia

Serum Sodium <120 mEq/L	Serum Sodium 120–130 mEq/L	Serum Sodium 130–135 mEq/L
• Headache • Restlessness • Lethargy • Seizures • Brain stem herniation • Respiratory arrest • Death	• Malaise • Unsteadiness • Headache • Nausea • Vomiting • Fatigue • Confusion • Anorexia • Muscle cramps	• Asymptomatic • Headache • Nausea • Vomiting • Fatigue • Confusion • Anorexia • Muscle cramps • Depressed reflexes

F. SIADH: Increased secretion of ADH (or vasopressin) results in increased water retention at the renal distal tubules.

Table 3. Typical Causes of SIADH

Neurologic & Pulmonary Disorders	Medications
TBI	Desmopressin/vasopressin
Brain tumor	Selective serotonin reuptake inhibitors
Stroke	Tricyclic antidepressants
Brain infection	Carbamazepine
Subarachnoid hemorrhage	Oxcarbazepine
Intracerebral hemorage	Chlorpropamide
Pneumonia/tuberculosis	Cyclophosphamide/iphosphamide
Lung cancer	Methylenedioxymethamphetamine
	Nicotine
	Opioids
	Phenothiazine antipsychotic medications
	Nonsteroidal anti-inflammatory drugs
	MDMA (ecstasy)

MDMA = 3,4-methylenedioxymethamphetamine; TBI = traumatic brain injury.

G. Treatment – Guidelines for diagnosing and managing hyponatremia are available (Intensive Care Med 2014;40:320-31).

Table 4. Treatment Strategies for SIADH

	Fluid Restriction	PO Sodium	IV Sodium	Demeclocycline	Vasopressin (V)-Antagonists
Mechanism of action	Restriction of free water results in increased impact of insensible losses, permitting sodium concentration to rise	Sodium supplementation	Sodium supplementation	Inhibition of ADH activity, possibly because of inhibition of aquaporin water channels	Inhibition of renal V2 vasopressin receptors (conivaptan V1A + V2 receptors; tolvaptan V2 only)

Table 4. Treatment Strategies for SIADH *(continued)*

	Fluid Restriction	PO Sodium	IV Sodium	Demeclocycline	Vasopressin (V)-Antagonists
Dose	< 1500 mL/day	4–16 g/day (1 g = 17 mEq of Na)	0.9%–3% at 0.5–1.5 mL/kg/hr	300 mg every 12 hr up to 1200 mg/day	Conivaptan: 20–40 mg IV daily Tolvaptan: 15–60 mg PO daily (Intensive Care Med 2014;40:320-31)
Efficacy	Modest, delayed (over 2+ days)	Modest, more effective for CSWS (over 2+ days)	Modest, more effective for CSWS (over 2+ days)	Modest, delayed (over 1 wk)	Modest, prompt over the first 24 hr
Common adverse effects	Thirst	Thirst, diarrhea, nausea, vomiting	Fluid overload, hyperchloremia	GI upset, hepatotoxicity, nephrotoxicity, photosensitivity	Thirst; conivaptan Infusion pain
Common considerations	Difficult to ensure adherence; caution for permitting hypovolemia in patients with cerebral perfusion needs such as SAH, TBI	Poor palatability when taken by mouth	≤ 3% sodium chloride may be given through peripheral IV line; The smallest-bore IV in the largest available vessel should be used. Central line remains the preferred access, if available	Chelation occurs with coadministered cations; nephrotoxicity has been reported	Cost; drug-drug interactions are common; risk of overcorrection of Na > 8 mEq/24 hours; tolvaptan should not be used > 30 days in patients with liver disease due to risk of life-threatening liver injury

ADH = antidiuretic hormone; IV = intravenous(ly); PO = oral(ly).

1. Treatment of SIADH can be challenging in neurocritical care patients such as those with SAH and TBI.
2. Treatment of choice is fluid restriction, which is typically not feasible in patients with SAH or TBI because of prioritizing a euvolemic volume status to optimize cerebral perfusion pressure (CPP). This is particularly important when treating elevated ICP or cerebral vasospasm.
3. Hypertonic sodium solutions are often needed to raise serum sodium.
4. A practice-based study evaluated the impact of various hyponatremia treatments on the serum sodium concentration (Neurocrit Care 2017;27:242-8). Hypertonic saline solutions were commonly used and most effectively increased the serum sodium.

H. Cerebral Salt-Wasting Syndrome (CSWS)
1. Etiology is largely unknown, but speculation typically focuses on the increased secretion of natriuretic peptides, causing loss of sodium at the renal distal tubules.
2. May also be associated with relative adrenal insufficiency
3. Typical causes include TBI, SAH, and brain tumor.

4. Fludrocortisone 0.1–0.4 mg/day in divided doses may help reduce sodium loss in CSWS (Arch Intern Med 2008;168:325-6).
    a. An adverse effect of fludrocortisone is hypokalemia: Consider potassium supplementation.

I. Considerations for Rapid Correction of Hyponatremia
1. Recommended increase in serum sodium concentration is 0.5 mEq/L/hour or less (Box 1).
2. Patients with chronic hyponatremia may need to be corrected more slowly because of the equilibration of brain electrolytes with chronic hyponatremic state (N Engl J Med 2000;342:1581-9).
3. Patients with acute hyponatremia may tolerate quicker correction.
4. Patients with severe neurologic symptoms because of hyponatremia appear to tolerate rapid correction of serum sodium. A strategy to rapidly correct sodium is to increase by 5 mEq/L in the first 1–2 hours, a maximum of 10 mEq/L in the first 5 hours, then 15–20 mEq/L in the first 48 hours (Crit Care Med 2017;45:1762-71).

Box 1. Common Equations Used for Sodium Correction in Hyponatremia

Na requirement (mmol) = total body water (0.6 × kg)[a] × (desired Na − current Na)
Infusion rate (mL/hr) = (Na requirement × 1000)/(infusion Na concentration × time)

[a]0.6 for men, 0.5 for women; infusion Na concentration in millimoles per liter and time in hours.
Na = sodium.
Information from: Adrogue HJ, Madias N. Hyponatremia. N Engl J Med 2000;342:1581-9.

5. Quicker correction in patients with severe symptoms (coma, seizures) may be prudent – Up to 1–2 mEq/L/hour for the first few hours
6. Rapid correction necessitates frequent serum sodium monitoring (e.g., every 4 hours) to avoid overcorrection or too-rapid correction.
7. Too-rapid correction of serum sodium may result in osmotic demyelination syndrome; a routine approach should be to limit Na increase to not more than 12 mEq/L in the first 24 hours or less than 18 mEq/L in the first 48 hours (Intensive Care Med 2014;40:320-31).

II. HYPERNATREMIA

A. Epidemiology
1. Hypernatremia (serum sodium greater than 150 mEq/L) is also common in patients with neurologic injury.
2. Incidence in SAH up to 22%
3. Incidence in TBI up to 21%
4. Hypernatremia and hyperchloremia are increasingly associated with an increased risk of acute kidney injury; thus, careful monitoring is warranted.

B. Diagnosis/Pathophysiology: Laboratory tests (serum sodium) are needed to diagnose hypernatremia. Urine sodium, urine osmolality (<250 mOsm/L), urine specific gravity (<1.004), urine volume, and serum osmolality (>300 mOsm/L) may also help determine the specific pathogenesis.

C. Typical Causes
   1. Consideration of iatrogenic hypernatremia
   2. Diabetes insipidus
      a. Decreased secretion of ADH or vasopressin results in decreased retention of water at the renal distal tubules. Observed in the setting of pituitary or hypothalmic damage (brain tumor, TBI) or brain death.
      b. Characterized by voluminous (greater than 250 mL/hour), dilute urinary output

D. Treatment
   1. Hypotonic solutions for free-water replacement
      a. Dextrose 5% in water
      b. 0.45% sodium chloride
      c. Water supplementation orally or by feeding tube
   2. Vasopressin analogs
      a. Supplementation of ADH to normal functional concentrations
      b. Titrate therapy to normalized urinary output, serum sodium correction, and urine specific gravity.
      c. Desmopressin
         i. Intravenously or subcutaneously: 0.5–4 mcg every 8–12 hours (usual starting dose 1–2 mcg)
         ii. Intranasally: 10–40 mcg/day divided into two or three doses (usual starting dose 10 mcg)
         iii. Orally: 50–800 mcg divided into two doses (usual starting dose 50 mcg)
         iv. May be dosed as needed, depending on initial laboratory values
      d. Patients after pituitary removal may more commonly require long-term therapy.
   3. Arginine vasopressin – Continuous infusion 1–15 units/hour (usual starting dose 1 unit/hour; titrate to urinary output)
   4. Considerations for rapid correction of hypernatremia
      a. Recommended decrease in serum sodium concentration is 0.5 mEq/L/hour or less.
      b. Too-rapid correction of serum sodium may result in cerebral edema.
      c. In general, neurocritical care patients should receive minimal amounts of dextrose or free water–containing fluids to avoid the risk of cerebral edema.

III. STATUS EPILEPTICUS

A. Epidemiology – Incidence of status epilepticus is 18.3–41/100,000 people per year in the United States

B. Status Epilepticus – Continuous seizures for 5 minutes or more OR intermittent seizures without regaining consciousness in between seizures (N Engl J Med 1990;323:497-502; Neurocrit Care 2012;17:3-23)
   1. Generalized convulsive status epilepticus: Generalized convulsions present with onset of seizure activity – Usually clinically evident
   2. Nonconvulsive status epilepticus: Typically occurs in comatose patients with no overt convulsions present (seizures noted with electroencephalogram [EEG] monitoring only)

C. Refractory Status Epilepticus (RSE) – Status epilepticus that persists after standard treatment (e.g., a benzodiazepine followed by another anticonvulsant medication)

D. Super-refractory status epilepticus – Failure to wean continuous anesthetic agent after 24–48 hours of burst suppression or termination of continuous EEG seizures

E. Diagnosis/Pathophysiology – Diagnostic tests
   1. Laboratory tests often show electrolyte abnormalities (particularly sodium, magnesium, and phosphorus), and serum lactate may be elevated in the acute period after seizure activity.
   2. EEG monitoring is necessary to identify and characterize seizures.
      a. Continuous monitoring is preferred in patients with status epilepticus to capture intermittent or fluctuating seizure patterns (Neurocrit Care 2012;17:3-23).
      b. Typical recommended duration is at least 48 hours, and monitoring should be initiated as soon as possible after suggestion or diagnosis of seizure.

F. Causes

Table 5. Typical Etiologies of Status Epilepticus

Cause of Status Epilepticus	Approximate % of Patients
Epilepsy	33–55
Miscellaneous	12–24
Stroke	14–22
Anticonvulsant nonadherence	20
Drug withdrawal	10–14
Brain tumor	10
Metabolic	10
TBI	7
Drug toxicity	5
CNS infection	3

CNS = central nervous system; TBI = traumatic brain injury.

G. Clinical Impact
   1. Mortality rate ranges from 9% (primarily in patients with preexisting epilepsy/anticonvulsant medication nonadherence) to 30% (in patients with a concomitant pathology such as TBI or stroke).
      a. Mortality in nonconvulsive status epilepticus is about double compared with overt seizures.
      b. Older adults have a higher mortality rate.
   2. Discharge disposition: 14%–18% of patients presenting to the emergency department in status epilepticus ultimately have a resultant neurologic deficit.

H. Agent Selection (Neurocrit Care 2012;17:3-23; Epilepsy Curr 2016;16:48-61): Note that the dosing regimens for first- and second-line therapies are for children and adults
   1. First-line therapy (emergent)
      a. Benzodiazepine therapy preferred; appropriate dosing is essential – Do NOT underdose first-line therapy
      b. Lorazepam 0.1 mg/kg intravenously (maximum 4 mg/dose) OR
      c. Midazolam 5–10 mg (or 0.2 mg/kg, maximum dose 10 mg) intramuscularly (preferred for patients with no intravenous access) OR
      d. Diazepam intravenously (0.15–0.2 mg/kg, maximum 10 mg/dose) or rectally (0.2–0.5 mg/kg, maximum 20 mg/dose) – Not recommended as first-line therapy for hospitalized patients because of short duration of seizure control

2. Second-line therapy (urgent): Initiate an anticonvulsant after benzodiazepine therapy if seizures persist or if a maintenance therapy needs to be initiated to prevent future seizures.
    a. In 2020, the ESETT trial (Lancet 2020;395:1217-24) demonstrated that any of the following three drugs, at the listed dose, can be used as a potential first choice for a second-line therapy in status epilepticus
    b. Valproate 40 mg/kg (maximum dose 3000 mg) intravenously
    c. (Fos)phenytoin 20 mg/kg intravenously. (The ESETT Trial used a maximum dose of 1500 mg for blinding purposes but this is not recommended based on the PK/PD of [Fos]phenytoin.)
    d. Levetiracetam 60 mg/kg (maximum dose 4500 mg) intraveously
3. Third-line therapy (refractory)
    a. If seizures persist after first- and second-line therapy, RSE should be treated aggressively and in a timely manner with continuous infusion anesthetic agents. The patient should be intubated before starting therapy.
    b. RSE agents should be titrated to burst suppression (the almost total elimination of electrical activity on EEG, except for short "bursts" of cortical activity [typically therapy is titrated to 2–5 bursts/minute]) or termination of electrographic seizure activity, depending on provider preference.
    c. Maintain established goal (burst suppression or termination of electrographic seizure activity) for 24–48 hours before trying to taper continuous agents. In addition, optimize maintenance anticonvulsant therapy before wean.
    d. Limited data exist supporting any one agent over another.
    e. Continuous infusion anesthetics for RSE titrated to target burst suppression
        i. Midazolam high-dose infusion 0.05–2 mg/kg/hour
        ii. Pentobarbital infusion (loading dose about 25 mg/kg total, continuous infusion 0.5–5 mg/kg/hour)
        iii. Propofol infusion 20–200 mcg/kg/minute
    f. Other options for RSE in nonintubated patients
        i. Can use any of the second-line agents listed above if they have not already been administered
        ii. Lacosamide 400 mg intravenously
        iii. Topiramate 200–400 mg orally/nasogastrically
        iv. Phenobarbital 20 mg/kg intravenously
        v. Perampanel 12 mg orally/nasogastrically
4. Super-refractory status epilepticus
    a. Treatment is typically reinitiation of continuous anesthetic agents used for RSE.
    b. Ketamine, an *N*-methyl-d-aspartate (NMDA) receptor antagonist, may also be considered. Ketamine may cause emergence psychosis and hallucinations, for which midazolam may help.
    c. Patients should have a workup for causes of super-refractory status epilepticus (e.g., autoimmune encephalitis including paraneoplastic syndromes and, infectious encephalitis).
5. Nonpharmacologic therapies if medical therapy fails
    a. Ketogenic diet: Limiting medications and nutrition to low-carbohydrate content; more likely to be successful in pediatric patients than in adult patients because of difficulty achieving ketosis in adults. Exact mechanism of action is unknown, but may increase sequestration of glutamate and decrease reactive oxygen species. Benefits typically occur within 1–3 months. Must modify diet targeting fat to carbohydrate plus protein in a 4:1 or 3:1 ratio
    b. Vagus nerve stimulation

I. Monitoring
   1. Continuous EEG monitoring is necessary for status epilepticus and RSE.
   2. Proactively monitor serum concentrations for agents such as phenytoin and valproic acid to ensure adequate concentrations and mitigate the risk of toxicity. Because of altered protein binding, obtaining a free level of highly bound medications is preferred.

Table 6. Characteristics of Agents for Status Epilepticus

Antiepileptic Drug	Dosing	Common Adverse Effects	Considerations
Lorazepam	0.1 mg/kg IV (slow IV push) (maximum 4 mg/dose); up to 8 mg total	Sedation, hypotension	IV formulation contains propylene glycol
Midazolam (intermittent)	0.2 mg/kg IM (maximum 10 mg/dose)	Sedation, hypotension	Short duration with IV bolus
Diazepam	0.15 mg/kg IV (slow IV push); typically up to 10 mg total 0.2 mg/kg rectally (20 mg maximum)	Sedation, hypotension	IV formulation contains propylene glycol; rectal administration may be considered
Fosphenytoin	20 mg PE/kg IV (not to exceed 150 mg PE/min); may also give IM	Hypotension, arrhythmia, hepatotoxicity	Several drug-drug interactions
Phenytoin	20 mg/kg IV (not to exceed 50 mg/min)	Hypotension, arrhythmia, phlebitis, purple glove syndrome, hepatotoxicity	Several drug-drug interactions, IV formulation contains propylene glycol and ethanol. Addition of parenteral phenytoin to dextrose and dextrose-containing solutions should be avoided because of lack of solubility and resultant precipitation
Valproic acid	40 mg/kg IV (not to exceed 6 mg/kg/min)	Hyperammonemia, thrombocytopenia, hepatotoxicity	Many drug-drug interactions, avoid in patients with a TBI
Levetiracetam	60 mg/kg IV (maximum 4.5 g)	Sedation/paradoxical excitation, irritability	Renally eliminated, limited drug-drug interactions
Lacosamide	400 mg IV (typically over 15–30 min)	Dizziness, bradyarrhythmia	Limited drug-drug interactions
Topiramate	PO/enteral loading dose of 300–800 mg, followed by a daily dose of 400–1000 mg in 2–3 doses	Metabolic acidosis, nephrolithiasis	No IV formulation available
Phenobarbital	20 mg/kg (not to exceed 100 mg/min)	Sedation, hypotension, respiratory depression	IV formulation contains propylene glycol

Table 6. Characteristics of Agents for Status Epilepticus *(continued)*

Antiepileptic Drug	Dosing	Common Adverse Effects	Considerations
Pentobarbital	10 mg/kg (typically over 15–30 min, depending on blood pressure); typically need additional 5 to 10 mg/kg boluses to full loading dose of 25–30 mg/kg; 1 to 3 mg/kg/hr infusion	Sedation, hypotension, respiratory depression, constipation, cardiac depression, immunosuppression	IV formulation contains propylene glycol; target is burst suppression; several drug-drug interactions
Midazolam high-dose infusion	0.05–2 mg/kg/hr	Sedation, hypotension, respiratory depression	Potential tachyphylaxis, target is burst suppression
Propofol	20–200 mcg/kg/min, titrate by 5 mcg/kg/min; use caution when administering doses > 80 mcg/kg/min for extended periods	Sedation, hypotension, respiratory depression, pancreatitis, propofol-related infusion syndrome	Target is burst suppression; provides 1.1 kcal/mL; hypertriglyceridemia at higher doses
Ketamine	0.5–10 mg/kg/hr	Excitation, hypertension, possible neurotoxicity, hallucinations	May be more effective in prolonged refractory status epilepticus. May cause hypotension in patients with decreased shock index; consider IV fluid volume with high-dose infusions

AES = American Epilepsy Society; BID = twice daily; IM = intramuscular(ly); NCS = Neurocritical Care Society; PE = phenytoin equivalents; QID = four times daily.

Brophy GM, Bell R, Claassen J, et al. Guidelines for the evaluation and management of status epilepticus. Neurocrit Care 2012;17:3-23; Glauser T, Shinnar S, Gloss D, et al. Evidence-based guideline: treatment of convulsive status epilepticus in children and adults: report of the Guideline Committee of the American Epilepsy Society. Epilepsy Curr 2016;16:48-61.

Patient Case

1. A 37-year-old man (weight 75 kg) is admitted to the emergency department with meningitis. On admission, his GCS score decreases from E3-M6-V1T to E1-M5-V1T over 10 minutes, and his nurse notices facial twitching. An EEG is ordered. Current medications include famotidine 20 mg intravenously every 12 hours, heparin 5000 units subcutaneously every 8 hours, docusate 100 mg nasogastrically twice daily, and a supplemental vitamin infusion for potential alcohol withdrawal. Which is the best acute therapy for this patient's suspected seizure activity?
   A. Fosphenytoin 1500 mg PE intravenously × 1.
   B. Valproic acid 1500 mg intravenously × 1.
   C. Midazolam 10 mg intramuscularly × 1.
   D. Levetiracetam 1 g intravenously × 1.

IV. CENTRAL NERVOUS SYSTEM INFECTION: INTRAVENTRICULAR ANTIBIOTIC ADMINISTRATION

A. Case Selection
   1. Recommended in adult patients with cerebral spinal fluid (CSF) shunt or ventriculostomy infections for difficult-to-eradicate pathogens or for patients who cannot undergo the surgical component of therapy
   2. May also be considered in patients receiving systemic therapy who have responded poorly to systemic antimicrobial therapy alone (Clin Inf Dis 2017;64:e34-65)
   3. Not recommended for neonatal or infant central nervous system (CNS) infection

B. Appropriate Dosing
   1. Intravenous plus intraventricular is recommended.
   2. Use preservative-free formulations.
   3. Do not use diluents containing dextrose, whenever possible due to potential osmolality concerns.
   4. Do not use medications known to lower the seizure threshold (e.g., β-lactams).
   5. Daily dosing is usually necessary; may need to adjust according to the amount of CSF drainage from external ventricular drain

Table 7. Various Antimicrobials and Doses for Intraventricular Administration

Antimicrobial	Daily Dose/Volume (adults)	Approximate Osmolality (mOsm/kg)	Common Adverse Effects
Vancomycin	10–20 mg/1 mL of NS	291	Headache, mental status changes, possible hyponatremia
Gentamicin	4–8 mg/1 mL of NS	293	Seizures
Tobramycin	4–8 mg/1 mL of NS	283	Seizures
Amikacin	30 mg/1 mL of NS	383	Seizures
Polymyxin B	5 mg/1 mL of NS	10	Hypotonia, seizures, meningeal inflammation
Colistimethate	10 mg/3 mL of NS	367	Meningeal inflammation
Amphotericin B deoxycholate	0.5 mg/3 mL of SWI	256 (in dextrose 5%)	Nausea, vomiting
Daptomycin	5 mg/2 mL of NS	364	Unknown

NS = normal saline; SWI = sterile water for injection.

Cook AM, Mieure KD, Owen RD, et al. Intracerebroventricular administration of medications. Pharmacotherapy 2009;29:832-45; Mueller SW, Kiser TH, Anderson TA, et al. Intraventricular daptomycin and intravenous linezolid for the treatment of external ventricular-drain-associated ventriculitis due to vancomycin-resistant *Enterococcus faecium*. Ann Pharmacother 2012;46:e35.

V. INTRACRANIAL PRESSURE TREATMENT

A. General Concepts
   1. Elevated ICP decreases tissue perfusion and tissue oxygenation and worsens neurologic outcome.
   2. Monro-Kellie doctrine: ICP equals cerebral blood volume (10%) plus CSF (10%) plus brain tissue (80%). Each therapy targeted at decreasing ICP acts on one or more of these components.
   3. Most practitioners use a stepwise approach to treating elevated ICP, including the following interventions:
      a. Head-of-bed elevation (30–45 degrees): Optimizes venous return from the brain, reducing venous pooling

b. Osmotherapy (mannitol or hypertonic saline)
c. Acute hyperventilation: Reduction in P_{CO_2} to around 32 mm Hg causes a compensatory vasoconstriction, which reduces cerebral blood volume (chronic hyperventilation should be avoided because of complications such as stroke).
d. Drainage of CSF by a ventriculostomy
e. Sedation with or without neuromuscular blockade (avoid benzodiazepines, when possible)
f. Maintenance of CPP at 60–70 mm Hg
g. Surgical decompression or hemicraniectomy (depending on the clinical scenario)
h. Pharmacologic coma (pentobarbital)
i. Hypothermia (33°C–36°C) – Clinicians should consider the risk-benefit of moderate hypothermia versus targeted temperature management for control of ICP.

B. Treatment Thresholds
1. Recommendations are to treat sustained ICP greater than 22 mm Hg as measured by external ventricular drain, intraparenchymal catheter, or bolt (Neurosurgery 2017;80:6-15). If subarachnoid hemorrhage, symptom-based dosing over ICP targets is suggested (Neurocrit Care 2020;32;647-66).
2. Specific ICP threshold may have interpatient variability.
3. CPP ideally within 60–70 mm Hg
   a. Targeting CPP greater than 80 mm Hg routinely is also associated with an increased incidence of acute respiratory distress syndrome and mortality in patients with a TBI; thus, patients must be selected carefully when targeting higher ranges of CPP (Crit Care Med 1999;27:2086-95).
   b. Osmotherapy

C. Comparison of Osmotherapy Agents

Table 8. Comparison of Osmotherapy Agents

	Mannitol	**Hypertonic Saline**
Mechanism of action	Acute increase in cerebral blood flow results in cerebral vasoconstriction (because of autoregulation), leading to decreased cerebral blood volume Increase in serum osmolality creates osmotic gradient to pull extracellular fluid from brain Osmotic diuretic	Acute increase in cerebral blood flow results in cerebral vasoconstriction (because of autoregulation), leading to decreased cerebral blood volume Increase in serum osmolality creates osmotic gradient to pull extracellular fluid from brain
Typical dose	0.5–1 g/kg over 15 min (0.2-micron filter) Up to 1.6 g/kg if acute herniation	3%: 2.5–5 mL/kg over 15 min 7.5%: 1–2 mL/kg over 15 min 23.4%: 30 mL over 15 min (may infuse over 5–10 minutes in acute herniation)
Monitoring values	Serum: Osmolality, BMP (Na, K, SCr, BUN, glucose), osmolar gap Urine: Urinary output	Serum: Osmolality, Na, SCr, K Urine: Urinary output
Adverse effects	Hyper/hyponatremia Hypokalemia Renal failure Hypovolemia Rebound cerebral edema (?)	Hypernatremia Hypokalemia Hyperchloremic acidosis Renal failure Osmotic demyelination syndrome (?)

1. Metabolic acidosis may occur after several hypertonic saline doses as the result of hyperchloremia.
    a. In patients with acidemia, which complicates ventilator management or other aspects of care, a combination of sodium chloride and sodium acetate may be considered to maintain the hyperosmolarity of the solution but reduce chloride provision (Crit Care Med 1999;26:440-6).
    b. Sodium bicarbonate 8.4% may also be considered for acute ICP elevations when other osmotherapy options are not immediately available, such as in patient care areas that do not typically care for neurologic ICU patients or patients with a TBI (Neurocrit Care 2010;13:24-8).
2. Monitoring osmolar changes with mannitol: The traditional serum osmolality threshold was 320 mOsm/L when using mannitol.
    a. Theory was that serum osmolality values greater than 320 were associated with renal dysfunction.
    b. Osmolar gap appears to be a more appropriate and accurate method of evaluating renal dysfunction risk with mannitol.
    c. Approximates the mannitol concentration
    d. Osmolar gaps greater than 55 mOsm/kg have been associated with renal dysfunction; however, many centers choose to target a goal osmolar gap less than 20–30 mOsm/kg prior to re-dosing mannitol.
    e. Calculation of osmolar gap (Box 2)

Box 2. Calculation of Osmolar Gap

$$\text{Osmolar gap} = \text{measured osmolality} - \text{estimated osmolality}$$
$$\text{Osmolar gap} = \text{measured osmolality} - [(2 \times \text{Na}) + (\text{BUN}/2.8) + (\text{glucose}/18)]$$

D. Sedation – Mechanism of action: Decreased systemic oxygen delivery needs; reduced coughing, reduced agitation, decreased cerebral metabolic rate ($CMRO_2$)
    1. Propofol is typically the preferred sedative – Quick onset, short acting, less accumulation with prolonged duration
        a. Patients with a TBI (and other neurologic injuries) require frequent, accurate neurologic examinations to evaluate the evolution of the neurologic injury.
        b. Hypotension risk may be harmful in specific patient types (e.g., aneurysmal SAH/vasospasm, TBI, SCI).
        c. In contrast to midazolam, propofol is advantageous because of its short half-life and relative lack of residual sedative effects.
    2. Benzodiazepines
        a. Not preferred because of long duration of action
        b. Also associated with delirium and cognitive impairment
        c. Potential for withdrawal effect, seizures
    3. Dexmedetomidine
        a. No specific considerations in existing guidelines for neurocritical care patients
        b. Hypotension risk may be harmful in specific patient types (e.g., aneurysmal SAH/vasospasm, TBI, SCI).
        c. May be particularly helpful in patients with paroxysmal sympathetic hyperactivity (PSH) ("storming") and facilitating extubation, managing alcohol withdrawal, and sedation requiring frequent neurologic assessments (DEXPRONE study; J Crit Care 2020;60:79-83)

E. Neuromuscular Blockade
    1. Mechanism of action: Decreased systemic oxygen delivery needs; reduced coughing
        a. Neuromuscular blockers have no intrinsic value for reducing ICP, but they may be helpful in select patients with specific issues that exacerbate ICP elevations.

i. Prevention of cough, ventilator dyssynchrony (both increase ICP)
ii. Control P_{CO_2} (increased P_{CO_2} may also increase ICP)
b. Prevention of shivering during therapeutic hypothermia or targeted temperature management
c. Reduces intrathoracic pressure
d. May be essential in patients requiring high positive end-expiratory pressure (increased intrathoracic pressure may increase ICP)

2. Various agents may be useful – Depends on patient organ function, prescriber preference
a. Vecuronium/Rocuronium (particularly if normal organ function)
b. Cisatracurium (particularly if end-organ dysfunction)
c. Avoid atracurium, if possible, because of hypotension risk. Laudanosine, a hepatically metabolized product of Hofmann elimination, is a CNS stimulant that can accumulate with prolonged use.

3. Monitor by train-of-four (goal 1-2/4 twitches with no clinical evidence of neuromuscular function [e.g., overbreathing the ventilator rate]).

F. Metabolic Suppression (pentobarbital)
1. Mechanism of action: Suppression of electrical activity in brain (i.e., "burst suppression") causes reduced cerebral metabolic rate of oxygen ($CMRO_2$).
2. Reduced $CMRO_2$ leads to decreased cerebral blood volume.
3. Pentobarbital sodium is usually used in the United States (thiopental in Europe).
4. Risks may outweigh benefit, at least for certain conditions such as large hemispheric infarction.
5. Pentobarbital is no more effective and is potentially more harmful for first-line therapy for ICP control than mannitol – Early studies led clinicians to begin using pentobarbital only in patients with refractory ICP elevations (Can J Neurosurg 1984;11:434-40).
6. Typical pentobarbital dosage
a. 25–30 mg/kg intravenous loading dose. Usually given as 10 mg/kg × 1 dose, followed by 5 mg/kg every hour × 3 or 4 doses to avoid hypotension with large bolus dose
b. 1- to 5-mg/kg/hour infusion after loading dose
7. Titration
a. Titrated to goal ICP (usually less than 20 mm Hg)
b. Burst suppression on continuous EEG (target usually is 2–5 bursts/minute) is a surrogate end point for need of additional pentobarbital doses.
c. A bolus dose is required concomitantly with an infusion titration because of its long half-life (24–48 hours) and rapid redistribution.
8. Monitoring
a. ICP
b. EEG and burst occurrence per minute
c. Serum concentrations do not correlate well with ICP response and should not be used to titrate infusion. May be useful when therapy has been discontinued as part of brain death examination (to rule out continued intoxication from pentobarbital)
d. Due to the risk of metabolic acidosis from propylene glycol accumulation, serum chemistry, pH, and osmolality should be monitored during prolonged treatment.
9. Adverse effects
a. Hypotension as the result of several different causes
i. Propylene glycol diluent
ii. Direct vasodilator
iii. Reduction in sympathetic tone because of metabolic suppression
iv. Cardiac depressant (particularly with high doses and duration greater than 96 hours)
b. Bradycardia

c. Decreased GI motility and ileus
   i. Difficulty with enteral nutrition
   ii. Caloric needs are usually around 80%–90% of basal energy needs, so a lower flow rate for enteral nutrition is permissible.
   iii. Ideally, would use an elemental or semi-elemental nutrition product because stooling is rare on pentobarbital infusion
d. Infection (particularly pneumonia)
e. Immunosuppression
f. Withdrawal seizures may occur.
g. Metabolic acidosis due to propylene glycol toxicity

Patient Case

2. A 25-year-old man (weight 80 kg) is admitted after a two-story fall from a ladder. The initial CT scan of his brain reveals a large right temporal subdural hematoma, an overlying skull fracture, and a left temporal contusion. His post-resuscitation GCS is E1-M4-V1T. An ICP monitor is placed with an opening pressure of 32 mm Hg, and CPP is 53 mm Hg. Serum laboratory values include Na 139 mEq/L, K 3.6 mEq/L, BUN 42 mg/dL, SCr 2.4 mg/dL, glucose 178 mg/dL, WBC 14.8 × 10^3 cells/mm^3, pH 7.46, and P_{CO_2} 34 mm Hg. Which is the best initial therapy for this patient's elevated ICP?
   A. Mannitol 20% 1 g/kg intravenously × 1.
   B. 23.4% sodium chloride 30 mL intravenously × 1.
   C. Pentobarbital 10 mg/kg intravenously × 1.
   D. Midazolam 10 mg intravenously × 1.

G. TBI Guidelines (J Neurotrauma 2007;24:S1-S95; Neurosurgery 2017;80:6-15). The fourth edition of these guidelines was published in 2017; these are "living guidelines" that will be updated continuously. Unfortunately, the newest guidelines reflect only recommendations with high levels of evidence. Therefore, the 2007 guidelines may be needed to review practical recommendations (based on all levels of evidence) for complications in patients with a TBI.
   1. Seizure prophylaxis
      a. Recommended as an option for prevention of early posttraumatic seizures (first 7 days after event)
      b. Phenytoin/fosphenytoin is the most commonly recommended agent (because of support for use from prospective clinical trials) (N Engl J Med 1990;323:497-502).
      c. Levetiracetam is also commonly used, despite a paucity of data (monitor for pronounced behavioral adverse effects in patients with neurologic injury).
      d. Valproic acid is as effective as phenytoin, but a trend toward increased mortality was observed in a prospective clinical trial; thus, it is usually avoided if possible (J Neurosurg 1999;91:593-600).
      e. Use of anticonvulsants for prevention of late seizures (after 7 days) has not been proven effective (not recommended).
   2. CPP modulation
      a. CPP = mean arterial pressure (MAP) minus ICP.
      b. Surrogate for global cerebral perfusion
      c. Recommended goal is 60–70 mm Hg.
      d. Ideal CPP may have interpatient variability because of the patient's medical history and unique characteristics of the TBI.
      e. Patients with refractory elevations in ICP or who have a history of poorly treated hypertension (right shift of autoregulatory curve) may require a higher CPP.

3. Fluid resuscitation with or without vasopressor therapy
    a. Norepinephrine or phenylephrine are the preferred vasopressors for this indication.
4. Supportive care – Venous thromboembolism (VTE) prophylaxis
    a. Patients with a TBI have an increased risk of VTE because of:
        i. TBI-related coagulopathy
        ii. Delay in initiation of pharmacologic VTE prophylaxis
        iii. Immobility
        iv. Concomitant injuries (in polytrauma)
    b. Mechanical prophylaxis should be initiated as soon as possible.
    c. Pharmacologic prophylaxis should be initiated after intracranial bleeding is stabilized.
        i. Typically, 24–48 hours after event
        ii. May depend on coagulopathy on admission, extension of bleeding on CT scan, and other factors
        iii. Unfractionated heparin (every 8 hours) or low-molecular-weight heparin may be used for pharmacologic prophylaxis. In patients with normal renal function, low-molecular-weight heparin is preferred in those with polytrauma, particularly long bone or pelvic fractures.

H. Nutrition Support
1. Initiating nutrition support within 48 hours improves immune competence and may improve neurologic outcome (Crit Care Med 1999;27:2525-31).
2. Gastric feeding is not well tolerated in patients with a TBI, particularly during the first 5–7 days and particularly in those with an elevated ICP (causes decreased gastric motility). Postpyloric feeding access should be established as soon as possible.
3. Metabolic needs are elevated after a TBI (typically proportional to the severity of injury).
    a. Patients with a TBI typically require 120%–160% of basal metabolic needs.
    b. Metabolic cart/direct calorimetry can be used to better evaluate caloric needs.

I. Prevention of Stress-Related Mucosal Bleeding
1. Patients with a TBI have an increased risk of stress-related mucosal bleeding.
    a. Hypotension associated with a TBI or trauma
    b. Hypersecretion of acid associated with neurologic injury (Cushing ulcers)
    c. Potential for coagulopathy
    d. Need for mechanical ventilation
2. Almost all patients with a severe TBI should receive prophylaxis for stress-related mucosal bleeding.
    a. Histamine-2 receptor antagonists (H_2RAs) have traditionally been the preferred agents.
    b. Proton pump inhibitors (PPIs) also raise gastric pH and permit hemostasis in areas of gastritis.
    c. Recent meta-analyses have suggested that PPIs are superior to H_2RAs, but a well-powered clinical trial has not been completed in the ICU or the neurologic ICU population.
    d. Which agent to select may depend on:
        i. Medications taken at home before admission
        ii. Presence of GI bleeding on admission
        iii. Risk of *Clostridioides difficile* infection

J. Glycemic Control
1. Hyperglycemia is associated with increased mortality in TBI.
2. Potential mechanisms
    a. Glucose toxicity in neurons
    b. Surrogate for severity of injury
    c. Exacerbation of cerebral edema

3. Avoid administering dextrose 5% and other hypotonic glucose-containing fluids.
4. Glycemic goals
    a. Prevent hyperglycemia (greater than 180 mg/dL)
    b. 140–180 mg/dL seems reasonable to avoid hypoglycemia.
    c. Caution should be used with glucose values in the low-normal range because of the risk of hypoglycemia.
    d. Hypoglycemia is associated with a worse outcome in TBI.
        i. Glucose obligate substrate for neurons
        ii. Threshold for glucose needs may be altered in TBI.
        iii. May increase seizure risk

K. Steroids
1. High-dose methylprednisolone plays no role in the treatment of inflammation or edema associated with TBI.
2. Large, prospective, randomized clinical trial showed increased mortality in steroid group compared with placebo (CRASH) (Lancet 2005;365:1957-9).

L. Pharmacokinetic Alterations
1. Altered volume of distribution: Patients with a TBI have increased volume of distribution because of the following:
    a. Fluid resuscitation
    b. Reduced plasma protein binding (particularly with albumin as a negative acute-phase reactant)
    c. Transient increased permeability of blood-brain barrier
2. Hepatic metabolism induction
    a. TBI increases hepatic metabolic capacity (extent to which is likely proportional to severity of injury).
    b. Results in more effective clearance of hepatically metabolized medications
    c. Increased dosing requirement for commonly used agents such as phenytoin, midazolam
    d. Induction subsides over time (usually 1–3 months, but varies by patient).
    e. Hypothermia during TBI may also reduce the induction of hepatic metabolism/cause metabolic rate of medications to be less than baseline.
3. Augmented renal clearance
    a. Increase in glomerular filtration rate (e.g., CrCL greater than 120 mL/min)
    b. Fluid resuscitation
    c. Increased endogenous catecholamines and glucocorticoid response
    d. Results in more effective clearance of renally eliminated medications
    e. Increased dosing requirement for commonly used agents such as vancomycin, aminoglycosides, β-lactams (J Trauma Acute Care Surg 2016;81:1115-21; Neurocrit Care 2015;23:374-9; Clin Pharmacokinet 2010;49:1-16; Pharmacotherapy 2019;39:346-54)
    f. Augmented renal clearance tends to subside over time (usually after first 7–10 days, but varies by patient).

VI. PAROXYSMAL SYMPATHETIC HYPERACTIVITY (I.E., "BRAIN STORMING")

A. Epidemiology
   1. About 8%–33% incidence in survivors of acquired brain injury
   2. Commonly associated with TBI (specifically diffuse axonal injury and high-burden parenchymal lesions), but may occur with other CNS insults

B. Diagnosis
   1. Typically, three or more symptoms (J Neurotrauma 2014;31:1515-20)
      a. Fever
      b. Tachycardia
      c. Hypertension
      d. Tachypnea
      e. Dyspnea
      f. Diaphoresis
      g. Muscle rigidity (posturing)
   2. Common triggers
      a. Pain
      b. Bladder distension
      c. Turning
      d. Tracheal suctioning
      e. Typically unprovoked (hence "paroxysmal")

C. Pathophysiology largely unknown, but thought to be caused by somatosympathetic activation and heightened activity of brain stem after brain injury

D. Paroxysmal Sympathetic Hyperactivity (PSH) Assessment Measure: A tool to determine the possibility of PSH at that specific time has two components – Diagnosis Likelihood Tool and Clinical Feature Scale (J Neurotrauma 2014;31:1515-20)

Table 9. Preventive and Abortive Therapies for Paroxysmal Sympathetic Hyperactivity

Preventive Therapies	Abortive Therapies
Baclofen IT (titrated according to patient response)	Baclofen IT (titrated according to patient response)
Bromocriptine 1.25 mg enterally BID (up to 40 mg/day)	Dantrolene 0.25–2 mg/kg IV every 6–12 hr
Clonidine 0.1–0.3 mg enterally three times daily	Dexmedetomidine 0.2–1.5 mcg/kg/hr IV[a]
Gabapentin 100–300 mg three times daily (up to 4800 mg/day)	Diazepam 5–10 mg IV PRN
	Fentanyl 25–100 mcg IV PRN
Propranolol 20–60 mg enterally every 4–6 hr	Morphine 2–8 mg IV PRN[a]
	Propranolol 1–3 mg IV every 4 hr[a]

[a]Most common treatments.

IT = intrathecal; PRN = as needed.

Rabinstein AA, Benarroch EE. Treatment of paroxysmal sympathetic hyperactivity. Curr Treat Options Neurol 2008;10:151-7; Thomas A, Greenwald BD. Paroxysmal sympathetic hyperactivity and clinical considerations for patients with acquired brain injuries: a narrative review. Am J Phys Med Rehabil 2018 Jun 22. [Epub ahead of print]

VII. ACUTE ISCHEMIC STROKE

A. Epidemiology
   1. Fifth leading cause of death and No. 1 cause of disability in the United States, with around 800,000 strokes in the United States annually
   2. 85% of strokes in the United States are ischemic.

B. Diagnosis/Pathogenesis
   1. Diagnostic tests
      a. Neurologic examination
      b. Vital signs
      c. NIH Stroke Scale (greater than 25 is severe, range 1–42)
      d. Imaging and other tests (Stroke 2019;50:e344-418)
      e. Noncontrast CT scan or magnetic resonance imaging (MRI) of the brain (to rule out bleeding)
      f. CT angiography (for patients with evidence of large vessel occlusion)
      g. CT or MRI perfusion and diffusion imaging may be considered for patients outside the thrombolysis window.
      h. Chest radiography (if lung disease is suspected)
      i. Lumbar puncture (if SAH is suspected and CT scan is negative for blood)
      j. EEG (if seizures are suspected)
   2. Laboratory tests
      a. Blood glucose
      b. INR, activated partial prothrombin time (consider thrombin time, anti–factor Xa [anti-Xa] activity for direct oral anticoagulants)
      c. Complete blood cell count (CBC)
      d. Tests for hypercoagulable state

C. Causes
   1. Cardioembolic (29.1%)
   2. Large-artery atherosclerosis (16.3%)
   3. Lacunar infarcts (15.9%)
   4. Unknown (36.1%)
   5. Other (2.6%)

D. Treatment Considerations (2019;50:e344-418)
   1. 1. Thrombolysis
      a. Alteplase 0.9 mg/kg (maximum 90 mg) within 4.5 hours of symptom onset; 10% of total dose as intravenous bolus, followed by 90% as 60-minute intravenous infusion.
      b. Tenecteplase 0.25 mg/kg intravenous bolus (maximum 25 mg) within 4.5 hours of symptom onset may also be considered for off-label use in patients with acute ischemic stroke because it was found to be noninferior to alteplase (Lancet 2022;400:161-9), and was found to be superior in patients who are also eligible for mechanical thrombectomy (NEJM 2018;378:1573-82).
      c. In patients with acute ischemic stroke who awake with stroke symptoms or have unclear time of onset more than 4.5 hours from last known well, or at baseline state can undergo MRI to help select those who can benefit from intravenous alteplase administration within 4.5 hours of stroke symptom recognition (NEJM 2018;379:611-22).

Table 10. Typical Inclusion/Exclusion Criteria for IV Alteplase and Tenecteplase for Ischemic Stroke[a]

Patient Selection Criteria	Patient History Excludes All Contraindications
Onset of symptoms < 4.5 hr from drug administration	Intracranial or intraspinal surgery within 3 mo
Baseline head CT excludes intracerebral hemorrhage (ICH) or other risk factors	Head trauma or stroke < 3 mo
Age ≥ 18 yr	Active internal or intracerebral bleeding
	Symptoms suggestive of SAH
Vital signs and laboratory values:	Any history of ICH
INR ≤ 1.7	Intracranial neoplasm, arteriovenous malformation, or aneurysm
Plt ≥ 100,000/mm^3	Arterial puncture at noncompressible site within 1 wk
Blood glucose > 50 mg/dL	Current SBP > 185 mm Hg or DBP > 110 mm Hg
Blood pressure control (SBP < 185 mm Hg, DBP < 110 mm Hg)	Current use of anticoagulant agents with evidence of elevated sensitive laboratory tests, including direct-acting oral anticoagulants and therapeutic doses of low-molecular-weight heparin
	Additional exclusion criteria for 3- to 4.5-hr window:
	NIHSS > 25
	Current treatment with PO anticoagulants (regardless of INR)
	Evidence of ischemic injury > 1/3 of MCA territory

[a]Exclusions are primarily based on the risk of systemic bleeding or hemorrhagic conversion of stroke.

DBP = diastolic blood pressure; MCA = middle cerebral artery; NIHSS = NIH Stroke Scale (score); SBP = systolic blood pressure; tPA = tissue plasminogen activator.

Hacke W, Kaste M, Bluhmki E, et al. Thrombolysis with alteplase 3 to 4.5 hours after acute ischemic stroke. N Engl J Med 2008;359:1317-29. Demaerschalk BM, Kleindorfer DO, Adeoye OM, et al. Scientific rationale for the inclusion and exclusion criteria for intravenous alteplase in acute ischemic stroke: a statement for healthcare professionals from the American Heart Association/American Stroke Association. Stroke 2016;47:581-641. Powers WJ, Rabinstein AA, Ackerson T, et al. 2018 guidelines for the early management of patients with acute ischemic stroke: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. Stroke 2018;49:e46-e99. Powers WJ, Rabinstein AA, Ackerson T, et al. Guidelines for the early management of patients with acute ischemic stroke: 2019 update to the 2018 guidelines for the early management of acute ischemic stroke: a guideline for healthcare professionals from the American Heart Association/American Stroke Association. Stroke 2019;50:e344-e418.

2. Permissive hypertension
    a. Reduction in blood pressure after thrombolysis or recanalization is reasonable within the first 24 hours after the onset of stroke.
    b. Reduce blood pressure cautiously to avoid hypotension or underperfusion of infarcted area (less than 15% blood pressure lowering).
    c. Resumption of home blood pressure medications is reasonable 24 hours after the onset of stroke.
    d. Recommended to treat blood pressure in patients who do not receive thrombolytics if it exceeds SBP greater than 220 mm Hg or diastolic blood pressure (DBP) greater than 120 mm Hg
3. Seizure prophylaxis. Use of anticonvulsant medications for seizure prophylaxis is not indicated after ischemic stroke.
4. Mechanical thrombectomy and neuroendovascular interventions for ischemic stroke
    a. Neuroendovascular devices may be used to remove or disrupt clot to facilitate recanalization.
    b. No difference in safety outcomes when comparing usual care with usual care plus mechanical thrombectomy
5. Many clinical trials support mechanical thrombectomy in patients with acute ischemic stroke.
    a. Increases the likelihood of independence or improved modified Rankin Scale score compared with standard therapy
    b. Does not increase the risk of intracerebral hemorrhage (ICH)
    c. Extends the treatment window to 24 hours after last known well for both patients receiving tissue plasminogen activator (tPA) within the therapeutic window and those who are ineligible for tPA if they meet selection criteria

6. Blood pressure management after thrombectomy
    a. A retrospective, multicenter study of blood pressure after thrombectomy suggested that higher blood pressures were associated with more complications and worse outcomes (Stroke 2019;50:2448-54).
    b. The cohort with an SBP of 121–140 mm Hg appeared best in this study.
    c. The optimal blood pressure after thrombectomy is not well defined and needs further, high-quality study.
7. Secondary prevention
    a. Initiating aspirin (325 mg × 1; then 81–325 mg/day), a high-intensity statin, and an intensive blood pressure regimen is necessary.
        i. Ideally, give as soon as feasible after the onset of stroke.
        ii. Aspirin should not be initiated within 24 hours of alteplase.
    b. Clopidogrel 75 mg daily is also an option in patients whose aspirin therapy has failed or who have an aspirin allergy.
    c. Aggrenox (dipyridamole/aspirin) and ticagrelor plus aspirin (less than 100 mg) are also effective alternatives to aspirin for secondary prevention.
    d. Control/modification of other disease states is often necessary.
    e. Hypertension: Threshold to treat high blood pressure is greater than 130/80 mm Hg; typical blood pressure goal is less than 130/80 mm Hg (Stroke 2021;52:e364-467).
    f. Atrial fibrillation
        i. Rate or rhythm control
        ii. Anticoagulation (warfarin, direct-acting oral anticoagulants)
        iii. Typically, anticoagulant therapy is delayed until at least 5–14 days after stroke to reduce the risk of hemorrhagic conversion.
        iv. Avoid using loading doses of warfarin.
    g. Carotid artery stenosis: Stent versus endarterectomy (usually for patients with greater than 70% blockage and/or clinically evident symptoms)
        i. Aspirin 81–325 mg daily after endarterectomy
        ii. Dual antiplatelet therapy should be given for 3 months if stent placement is required, followed by one antiplatelet agent thereafter.
    h. Control of diabetes
    i. Identification and treatment of inherited or acquired hypercoagulable states

VIII. INTRACEREBRAL HEMORRHAGE

A. Epidemiology. Around 50,000 cases in the United States annually
    1. Diagnosis/pathogenesis
        a. Neurologic examination
        b. Vital signs
        c. NIH Stroke Scale and/or GCS score
    2. Imaging and other tests
        a. CT or MRI scan of the brain
        b. CT angiography or contrast-enhanced CT (to help identify patients at risk of hematoma expansion and to evaluate for underlying structural lesions)
        c. Medication history to identify agents that might produce coagulopathy
        d. Laboratory tests

e. Blood glucose
f. INR
g. CBC

B. Causes
1. Chronic/poorly treated hypertension
2. Oral anticoagulant use
3. Cocaine/other stimulant use
4. Ischemic stroke with hemorrhagic transformation
5. Chronic alcohol intake
6. Brain tumor
7. Arteriovenous malformation
8. Amyloid angiopathy

C. Clinical Impact – Death or major disability occurs in around 50% of patients.

D. Treatment Considerations (Stroke 2015;46:2032-60)
1. Anticoagulant reversal
a. Prompt anticoagulant reversal is necessary. Reversal of laboratory values does not mean that hemostasis will occur in all cases, especially with the newer oral anticoagulants (Table 11).
b. rFVIIa is not recommended for reversal of anticoagulation because it only replaces one clotting factor and does not appear to be effective for dabigatran reversal. However, if no other reversal agents are available, rFVIIa may be an option for patients who cannot accept human blood products.
2. Reversal of antiplatelet agents is somewhat controversial.
a. Two prospective studies suggest that platelet transfusion for patients taking antiplatelet agents with a new ICH is harmful (Lancet 2016;387:2605-13; J Neurosurgery 2013;118:94-103).
i. Both studies included patients with a small, stable ICH. Primarily basal ganglia hemorrhages, which are traditionally nonoperative
ii. Patients receiving dual antiplatelet therapy were underrepresented in both studies.
b. Routine administration of platelet transfusion for any patient with ICH who took antiplatelet agents before admission is not advisable.
c. Platelet transfusion is likely still indicated for patients who:
i. Have an urgent need for surgery or a ventriculostomy and have abnormal platelet function tests (if available)
ii. Have a large ICH, receiving antiplatelet agents and requiring neurosurgery
iii. Have acute neurologic decline with an intracranial hemorrhage

Table 11. Anticoagulant Reversal Options

Anticoagulant	Reversal Agent and Dose	Adverse Effects
Warfarin	4F-PCC [INR < 4 (25 units/kg, max 2500 units), INR 4–6 (35 units/kg, max 3500 units), INR > 6 (50 units/kg, max 5000 units)] + vitamin K 10 mg IV Alternative: FFP 10–15 mL/kg + vitamin K 10 mg IV	Thrombosis, anaphylactoid reaction, (vitamin K), pulmonary edema, or transfusion-related reaction (FFP), contains heparin (4F-PCC)
PO factor Xa inhibitors (rivaroxaban, apixaban)	Andexanet alfa (Andexxa) (see dosing information that follows in Table 12) or 4F-PCC 25–50 units/kg or activated 4F-PCC (FEIBA) 25–50 units/kg	Andexanet: Rebound anti-Xa therapeutic concentrations within 2 hr of infusion completion Thrombosis; limited data for reversal, especially with FEIBA
PO DTIs[a] (dabigatran)	Idarucizumab (Praxbind) 5 g IV × 1; repeat dose may be necessary in patients with high DTI exposure or poor renal function	Idarucizumab approved in late 2015 – antibody specifically for dabigatran Thrombosis; limited data analyses show minimal efficacy for reversal with PCC and FEIBA
PO antiplatelets	Platelet transfusion infusion (usefulness is uncertain – see Neurocrit Care 2016;24:6-45) Consider desmopressin 0.4 mcg/kg IV	Pulmonary edema or transfusion-related reaction
Unfractionated heparin	Protamine 1 mg (max 50 mg) for each 100 units of heparin (infused within the past 2–3 hr)	Hypotension, hypersensitivity
Low-molecular-weight heparins	Protamine 1 mg (max 50 mg) for each 1 mg of enoxaparin (within 8 hr of last dose)	Hypotension, hypersensitivity

[a]Reversal options have not been tested in patients with ICH and have had varying degrees of coagulopathy reversal in experimental animal and human models.

DTI = direct thrombin inhibitor; FEIBA = factor eight inhibitor bypassing activity; FFP = fresh frozen plasma; 4F-PCC = 4-factor prothrombin complex concentrate.

Table 12. Andexanet Alfa Dosing Based on Factor Xa Inhibitor Dose

Factor Xa Inhibitor	Last Dose	< 8 hr or Unknown Since Last Dose	≥ 8 hr Since Last Dose
Apixaban	≤ 5 mg	Low dose	Low dose
	> 5 mg or unknown	High dose	
Rivaroxaban	≤ 10 mg	Low dose	
	> 10 mg or unknown	High dose	

Low dose = 400 mg at a target rate of 30 mg/min, then 4 mg/min for up to 120 min; high dose = 800 mg at a target rate of 30 mg/min, then 8 mg/min for up to 120 min.

3. Blood Pressure Management
   a. Prompt control of blood pressure is essential.
   b. Historically, caution may have been used in rapidly reducing blood pressure in patients with chronic hypertension because of concerns regarding accommodations in cerebral autoregulation.
   c. However, there may be some benefit of rapidly reducing the blood pressure to approximately 140-150 mm Hg (and thus reducing the risk of hemorrhage expansion). This may outweigh concern for cerebral autoregulation issues and potential for ischemia, but patients with initial SBP >180 mm Hg may have a greater risk of renal dysfunction.
   d. INTERACT-2 – Large, prospective, randomized trial that compared blood pressure control within 1 hour (N Engl J Med 2013;368:2355-65). SBP less than 140 mm Hg was as safe and effective as SBP less than 180 mm Hg (and may have improved functional outcomes).

e. ATACH-2 – Large, prospective, randomized trial comparing blood pressure control within 4½ hours in patients with SBP >180 mm Hg. Targeting an SBP goal of 110–139 mm Hg was as effective as targeting an SBP goal of 140–179 mm Hg but was associated with increased renal adverse events at 3 months. (Mean minimum SBP in the first 2 hours after randomization was 128.9 mm Hg in the intensive-treatment group and 141.1 mm Hg in the standard-treatment group.) (N Engl J Med 2016;375:1033-43).

f. Evidence also suggests that blood pressure fluctuations in the first 4–6 hours post-ICH are associated with poor functional outcomes (Stroke 2018;49:348-54).

4. Seizure Prophylaxis. Use of anticonvulsants for seizure prophylaxis is not indicated after ICH.
5. VTE prophylaxis should be initiated about 24 hours after symptom onset or surgical evacuation.

Patient Case

Questions 3 and 4 pertain to the following case.
A 61-year-old man is admitted with acute onset of difficulty speaking, confusion, and right-sided weakness. His NIH Stroke Scale score is 20. A head CT scan reveals a right parietal intracerebral hemorrhage (ICH). The patient's home medications include hydroxychloroquine, ibuprofen as needed, warfarin, amlodipine, and donepezil. His medical history includes a deep venous thrombosis (1 year ago), hypertension, early dementia, and arthritis. Serum laboratory values include Na 140 mEq/L, K 3.6 mEq/L, BUN 27 mg/dL, SCr 1.8 mg/dL, glucose 289 mg/dL, Hct 36.7%, Plt 245,000/mm^3, and INR 6.8. His vital signs include blood pressure 163/101 mm Hg, heart rate 99 beats/minute, Sa_{O_2} 97%, and respiratory rate 20 breaths/minute.

3. Which is the most appropriate initial therapy in addition to vitamin K for this patient's care?
   A. Reinitiate amlodipine.
   B. Give 4-factor prothrombin complex concentrate (4F-PCC) 50 units/kg intravenously × 1.
   C. Give 6-pack infusion of platelets.
   D. Give recombinant factor VIIa (rFVIIa) 90 mcg/kg intravenously × 1.

4. For this 61-year-old patient with ICH, which is the most appropriate initial antihypertensive therapy?
   A. Clevidipine 1-mg/hour infusion to keep SBP 180 - 220 mm Hg.
   B. Nicardipine 5-mg/hour infusion to keep SBP 140 - 150 mm Hg.
   C. Labetalol 10 mg intravenously as needed to keep SBP 160 - 200 mm Hg.
   D. Esmolol 50-mcg/kg/minute infusion to keep SBP 120 - 130 mm Hg.

IX. SUBARACHNOID HEMORRHAGE

A. Epidemiology
   1. Occurs in around 11.4 per 100,000 person-years. Worldwide incidence is around 6.1 per 100,000 person-years.
   2. 60%–70% female, typical age 40–60 years

B. Diagnosis/Pathogenesis
   1. Serial neurologic examination
   2. Vital signs
   3. NIH Stroke Scale and/or GCS score

Table 13. SAH Severity Scale Scores

	Score Range	Comments
Hunt and Hess	0 (no symptoms) to 5 (moribund)	Best correlated with risk of mortality
World Federation of Neurological Surgeons	1 (GCS score 15, no deficit) to 5 (GCS score 3–6)	Integrates risk of mortality and motor dysfunction
Fisher	1 (no blood visualized) to 4 (diffuse SAH, ICH, or intraventricular hemorrhage present)	Best correlated with risk of vasospasm

C. Imaging (Stroke 2023;54:314-70)
   1. CT scan of brain
   2. Lumbar puncture when CT scan of brain is negative for blood
   3. Digital subtraction ("conventional") angiography
   4. May use CT angiography or magnetic resonance angiography if conventional angiography is unavailable
   5. Transcranial Doppler, often daily during peak vasospasm risk period

D. Medication History – To identify agents that might produce coagulopathy or hypertension

E. Laboratory and Other Tests
   1. INR
   2. CBC
   3. Troponin
   4. ECG (electrocardiogram)
   5. Echocardiogram

F. Causes – Typically caused by cerebral aneurysm. Modifiable risk factors for SAH:
   1. Hypertension
   2. Smoking
   3. Illicit drug use

G. Clinical Impact
   1. Sudden death: Around 22%–26% of patients die before hospitalization. Inpatient mortality 13.1% in 2018 (United States).
   2. Vasospasm and delayed ischemic neurologic deficits
      a. Presence of blood in subarachnoid space elicits a chemical meningitis-type inflammatory response and results in hemolysis of subarachnoid blood.
      b. Vasospasm (persistent vasoconstriction) occurs, reducing distal cerebral blood flow.
         i. Typical course is 3–14 days.
         ii. Vasospasm risk peaks at around 7–10 days.
      c. Several mechanisms of pathogenesis
         i. Inflammatory infiltration
         ii. Endothelin activation
         iii. Liberation of hemoglobin results in the scavenging of nitric oxide.
      d. Vasospasm is one of the main factors resulting in death or disability after an acute SAH, aside from initial ictus.

H. Treatment Considerations (Stroke 2023;54:314-70)
   1. Surgical management to prevent rebleeding and 'secure' aneurysm
      a. Craniotomy and aneurysm clipping
      b. Endovascular aneurysm coiling
   2. Agents for preventing vasospasm or delayed ischemic neurologic deficits
      a. Nimodipine (Br Med J 1989;298:636-42)
         i. Is a "cerebrovascular-specific" lipophilic dihydropyridine calcium channel blocker
         ii. 60 mg orally or by tube every 4 hours × 21 days (dose reduction 30 mg every 2 hours may be considered if 60 mg dose intolerable): The clinical impact of dose reducing or discontinuing nimodipine because of hypotension is not well understood but may be associated with unfavorable outcomes (Neurocrit Care 2016;25:29-39).
         iii. Only U.S. Food and Drug Administration (FDA) label-approved medication to reduce delayed ischemic neurologic deficits associated with SAH
         iv. Clinical trials did not show a large effect of nimodipine on the occurrence of vasospasm (though the incidence of delayed ischemic neurologic deficits were significantly less).
         v. Possibly neuroprotective
         vi. Administration issues in patients who require enteral doses – Black box warning against inadvertent intravenous administration
         vii. Nursing should be prohibited from extracting the gel inside the nimodipine capsule for bedside administration (may increase the risk of inadvertent intravenous administration, incomplete extraction from the capsule) (Neurocrit Care 2015;22:89-92).
         viii. There is a commercially available liquid product that is for use in patients with swallowing difficulty or feeding tubes. Pharmacy compounding of nimodipine syringes from liquid capsules has been reported but further studies are warranted to determine the optimal enteral formulation. (Pharmacotherapy 2023;43:279-90).
      b. Antifibrinolytic agents
         i. Use of antifibrinolytic agents such as aminocaproic acid or tranexamic acid has been evaluated after SAH.
         ii. Older data analyses before the advent of endovascular interventions suggested that rebleeding is less common with these agents (primarily aminocaproic acid) but that stroke may be more common.
         iii. More recent data analyses suggest that a short infusion (less than 72 hours) reduces rebleeding but probably does not affect long-term outcomes.
            (a) The 2023 American Heart Association SAH guidelines moderately support this strategy in early SAH management (Stroke 2023;54:314-70)
            (b) For example, use of 1000 mg of tranexamic acid intravenously every 6 hours until the aneurysm is secured
      c. Statins
         i. Preservation of nitric oxide balance as heme is liberated during SAH hemolysis.
         ii. Phase II data with pravastatin and simvastatin
         iii. Phase III trial (STASH) showed no benefit of applying statins in aneurysmal SAH (Lancet Neurol 2014;13:666-75).
         iv. Abrupt withdrawal of statins in patients who were taking statins before SAH may result in a withdrawal effect and increase the risk of vasospasm.
      d. Other studied agents
         i. Magnesium – No usefulness in attaining magnesium concentrations 3–4 mEq/L. Maintaining magnesium at normal concentrations (i.e., preventing hypomagnesemia) is advisable.
         ii. Clazosentan – No usefulness in blocking endothelin-1
         iii. Albumin – Not beneficial and may be associated with an increase in pulmonary edema

3. Treatment of vasospasm
    a. Intra-arterial therapies (see Table 14)
    b. Triple-H therapy (hypertension, hypervolemia, hemodilution)
        i. No longer recommended in the traditional format. Euvolemia is better than hypervolemia – Similar outcomes in clinical trials, less pulmonary edema
        ii. Hemodilution has not been shown beneficial.
    c. Hyperperfusion therapy
        i. Better descriptor for the goal of therapy
        ii. Vasospasm causes distal vasoconstriction to the point of ischemia.
        iii. Maximizing cerebral blood flow mitigates ischemia.
    d. Contemporary therapy includes:
        i. Euvolemia
        ii. Vasopressors (blood pressure targets are ill defined, but are also typically patient- and symptom-dependent)
        iii. Inotropes (milrinone)
        iv. Superselective intra-arterial vasodilators (e.g., verapamil, nicardipine)
4. Headache
    a. "Worst headache of my life" is the typical chief concern for an aneurysmal SAH.
    b. Multimodality treatment will be needed to improve patient satisfaction during hospital admission (which may be up to 14–21 days).
    c. Optimal choice of agents is not well known.
        i. Common analgesics such as acetaminophen, nonsteroidal anti-inflammatory drugs (if aneurysm is secured)
        ii. Other therapies such as magnesium, corticosteroids, or gabapentin may be helpful.
        iii. Agents that contain caffeine are often used for headache but may not be appropriate, given the risk of cerebral vasospasm after SAH.
        iv. Opioids may be needed, but use should be limited to avoid sedation and inability to monitor neurologic examination.
        v. Nonpharmacologic therapies like warm compresses on the back of the neck, minimization of noise, and minimization of bright light may also be useful.
5. Seizure prophylaxis (Stroke 2023;54:314-70)
    a. Use of anticonvulsants for seizure prophylaxis is controversial after SAH.
    b. For new-onset seizures after aSAH, treatment with antiseizure medication for 7 days is recommended.
    c. Prophylactic antiseizure medication should not be routinely used but can be considered in high-risk patients (with ruptured middle cerebral artery aneurysm, ICH, high-grade aSAH, hydrocephalus, or cortical infarction).
    d. Phenytoin use is associated with excess morbidity and should be avoided.
6. Role of nicotine replacement therapy (NRT) – Smoking is a common risk factor for developing cerebral aneurysms.
    a. Nicotine administration to naive users causes vasoconstriction, whereas administration to heavy users causes little reaction. Nicotine withdrawal in heavy users causes acute, temporary vasodilation, which may lead to headaches.
    b. Seder et al.: Neurocrit Care 2011;14:77-83: Evaluated safety of NRT in active cigarette smokers admitted with an aneurysmal SAH and included 128 who received NRT and 106 who did not. Despite its vasoactive properties, NRT administered among active smokers with acute aneurysmal SAH appeared safe, with similar rates of vasospasm and delayed cerebral ischemia and a slightly higher rate of seizures.

c. Panos et al.: Am J Health Syst Pharm 2010;67:1357-61: No difference in unfavorable discharge disposition among neurosurgery ICU patients who were smokers treated with NRT (n=114), smokers not treated with NRT (n=113), and nonsmokers not treated with NRT (n=113). Primary admitting diagnosis for neurosurgery patients included SAH, ICH, other trauma, and elective neurosurgery.

X. INTERVENTIONAL ENDOVASCULAR MANAGEMENT

A. Intra-arterial Therapies (Pharmacotherapy 2010;30:405-17)
   1. Administration technique -- typically administered during cerebral angiography
   2. Catheter advanced to vessels with lesion/vasospasm, and drug is infused locally ("superselective infusion")

B. Vasodilators
   1. Calcium channel blockers
      a. Typically used for cerebral vasospasm associated with SAH
      b. Direct, local infusion typically results in immediate vasodilation.
      c. Usually effective in proximal and distal vessels
   2. A trial is ongoing to determine the optimal intra-arterial drug treatment regimen (nicardipine vs. verapamil vs. nicardipine plus verapamil plus nitroglycerin) for arterial lumen restoration post-cerebral vasospasm after aneurysmal SAH (Intra-arterial Vasospasm Trial at https://clinicaltrials.gov/ct2/show/record/NCT01996436).

Table 14. Typical Agents for Intra-arterial Use for Cerebral Vasospasm

Agent	Typical Dose	Adverse Effects
Nicardipine	2–25 mg, up to 5 mg/vessel	Systemic hypotension Increased ICP
Verapamil	1–10 mg	Systemic hypotension Bradycardia Increased ICP
Milrinone	5–15 mg	Systemic hypotension

ICP = intracranial pressure.

C. Intra-arterial Thrombolysis
   1. Most often used in patients with ischemic stroke
   2. Limited evidence to support combining with intravenous thrombolytic as a standard of care
   3. Current roles
      a. Combination with mechanical thrombectomy
      b. Rescue therapy in patients having received intravenous thrombolytic
      c. Large hemispheric infarction
         i. Dose is not well defined.
         ii. Typically applied until thrombus has resolved
         iii. Alteplase dose less than 20 mg

Patient Case

Questions 5 and 6 pertain to the following case.
A 49-year-old woman presents to an urgent treatment center with the "worst headache of her life." She is sent to your emergency department, where a head CT scan reveals a diffuse SAH. The patient takes no home medications and has an insignificant medical history other than a 20 pack-year history of smoking.

5. Which therapy is most appropriate to prevent ischemic complications from SAH?
   A. Nimodipine for 21 days.
   B. Euvolemia and permissive hypertension for 14 days.
   C. Simvastatin for 14 days.
   D. Aminocaproic acid infusion for 48 hours.

6. On hospital day 5, the patient has reduced alertness, and her GCS score decreases by 2 points. The digital subtraction angiography suggests cerebral vasospasm. Which treatment modality is best to initiate first?
   A. Norepinephrine 0.05 mcg/kg/minute to increase MAP to 90 mm Hg.
   B. One unit of packed RBCs to increase Hgb to 10 g/dL.
   C. 3% sodium chloride boluses to increase central venous pressure to 14 mm Hg.
   D. Milrinone 0.375-mcg/kg/minute infusion to increase cardiac index to 5 L/minute/m^2.

D. Stent Deployment and Antiplatelet Agents
   1. Intracranial stents are often deployed in place of coils or to support coils for complex aneurysms.
   2. Intracranial circulation is different from coronary circulation.
      a. Blood vessels are generally smaller and more tortuous.
      b. Flow rate is lower.
      c. Epithelialization of stent takes longer.
   3. Dual antiplatelet therapy is typically used around the time of stent placement.
      a. Clopidogrel (a prodrug) plus aspirin
      b. Current evidence suggests therapy for up to 3 months (not 4 weeks like after percutaneous coronary intervention), followed by aspirin monotherapy thereafter.
      c. Platelet testing may be necessary in some individuals to evaluate their pharmacogenetic response to clopidogrel.
      d. No gold standard for platelet reactivity testing in this setting. VerifyNow is commonly used and measures platelet reactivity units (PRU) for aspirin and P_2Y_{12} inhibitors.
      e. Some patients may not respond to clopidogrel (up to 30%), and additional loading doses and increased maintenance doses are not effective.
      f. Ticagrelor may play a role in patients with a variable response to clopidogrel (must be used in combination with aspirin less than 100 mg) (Neurocrit Care 2024;40:262-71).
      g. Prasugrel (a prodrug) is not recommended (unless absolutely necessary) in patients with high on-treatment platelet reactivity (e.g. clopidogrel non-responders) because of warnings about use in patients with a history of stroke – Increased bleeding. Reduced doses (2.5–5 mg) have been reported with some success in preventing thrombosis without excessive bleeding.

Patient Case

Questions 7 and 8 pertain to the following case.
A 52-year-old woman is admitted to your ICU after a single-vehicle crash. She has many orthopedic injuries to her lower extremities and a subdural hematoma, which will require an emergency craniotomy for evacuation. Her home medications include metoprolol, rivaroxaban 10 mg (last dose 5 hours ago), lisinopril, and atorvastatin. Her medical history is significant for atrial fibrillation. Laboratory values on admission were notable for Hct 31.2% and Plt 577,000/mm^3.

7. Which laboratory test would best evaluate the extent of anticoagulation from rivaroxaban?
   A. INR.
   B. Anti-Xa activity concentration.
   C. Activated partial thromboplastin time.
   D. VerifyNow PRU test measurement.

8. Which therapy would be most appropriate to reverse rivaroxaban before this patient's emergency craniotomy?
   A. 4F-PCC 50 units/kg intravenously × 1.
   B. Andexanet 200 mg intravenously x 1.
   C. Fresh frozen plasma 15 mL/kg intravenously × 1.
   D. rFVIIa 90 mcg/kg intravenously × 1.

XI. ACUTE SPINAL CORD INJURY (NEUROSURGERY 2013;60(SUPPL 1):82-91)

A. Epidemiology – Annual incidence of 15–40 cases per 1 million people in the United States

B. Diagnosis/Pathogenesis
   1. Diagnostic tests
   2. Neurologic examination

C. Imaging (Neurosurgery 2013;60(suppl 1):82-91)
   1. CT scan of spine
   2. Many views of spine radiography are necessary when a CT scan is unavailable.

D. Causes
   1. 40%–50% are caused by motor vehicle collisions.
   2. Falls (20%), violence (14%), recreational and work activities

E. Clinical Impact
   1. Mortality
      a. 50%–75% at the time of injury
      b. Hospital mortality 4.4%–16%
   2. Morbidity
      a. Paralysis and loss of sensation
      b. Spasticity

c. Orthostatic hypotension
d. Autonomic dysreflexia
e. VTE
f. Decubitus ulcers
g. Respiratory insufficiency
h. Bowel and bladder dysfunction
i. Sexual dysfunction
j. Treatment considerations
k. Neurogenic shock
    i. Hypotension often occurs after injury (50%–90% of cervical spine injuries).
    ii. May be associated with malperfusion of the spinal cord and worsened outcomes
    iii. Etiology of shock is decreased sympathetic nervous system outflow. Continues to be counterbalanced by parasympathetic outflow, which is not affected by SCI
    iv. Results in hypotension and bradycardia

F. Blood Pressure Management
1. Typical recommendations after an acute SCI are to maintain MAP 85–90 mm Hg for 7 days to ensure adequate spinal perfusion.
2. Little high-quality evidence supports this recommendation, but the recommendation is included in the SCI guidelines as a treatment option.
3. Often requires judicious fluid resuscitation and vasopressor support
4. Persistent hypotension may be treated with fludrocortisone or midodrine.
5. Persistent bradycardia may be treated with pseudoephedrine or low-dose theophylline.
6. Droxidopa, an enteral analog of norepinephrine, may be useful for blood pressure augmentation in this setting.

G. VTE Prophylaxis
1. VTE occurs in 80%–100% of patients without pharmacologic prophylaxis
2. Low-molecular-weight heparins are the drugs of choice for prophylaxis and should be initiated within 36 hours post-injury.
3. Duration of prophylaxis is typically about 8 weeks.

H. Role of High-Dose Methylprednisolone (Neurosurgery 2013;60(suppl 1)82-91; J Neurosurg 1998;89:699-706; JAMA 1997;277:1597-604; J Neurosurg 1992;76:23-31; N Engl J Med 1990;322:1405-11)
1. Controversial topic related to NASCIS-II and NASCIS-III trials
2. Methylprednisolone 30 mg/kg intravenously × 1, followed by 5.4 mg/kg/hour within 8 hours of injury for 24 or 48 hours
3. Both trials suggested a modest benefit in the first 6 weeks or 6 months (which often did not persist at 1 year) and a modest risk (primarily related to infection). Current guidelines do not support administering high-dose methylprednisolone.
4. NASCIS-II split enrolled population in half (those who received the drug before the median time to administration [8 hours] and those who did not).
    a. Subgroup analysis may not have been powered to show benefit.
    b. Reported motor and sensory scores from one side of the body, not both. The investigators later said there was no difference but have not allowed others to examine the raw data.
5. Consistently showed risk (GI bleeding, infection) and inconsistently showed benefit
6. Potential treatment effects may have been caused by early surgery or additional benefit of high-dose methylprednisolone therapy in combination with early surgery.

7. NASCIS-III used a functional independence measure (FIM) score to show how improvement in muscle strength might translate to improved outcomes. Failed to show a difference in FIM score
8. If a practitioner does choose to use high-dose steroids in SCI:
    a. Must use methylprednisolone; no other steroids
    b. Must use NASCIS-II or NASCIS-III dosing
    c. Must give within 8 hours of injury

XII. BRAIN TUMORS

A. Epidemiology
    1. According to the National Brain Tumor Society, an estimated 94,000 Americans received a new primary brain tumor diagnosis in 2023. Primary brain tumors (from brain cells such as meninges and neural tissues)
        a. Glioblastoma
        b. Meningioma
        c. Pituitary adenoma
        d. Astrocytoma
    2. Brain metastases – Common neoplasms that result in spread to brain
        a. Lung (40%–50%)
        b. Breast (15%–20%)
        c. Melanoma (5%–10%)
        d. Colon (4%–6%)
        e. Renal cell carcinoma
        f. CNS lymphoma

B. Diagnosis/Pathogenesis – Diagnostic tests
    1. Contrast-enhanced MRI is the most common test.
    2. Biopsy is often necessary to reveal the specific histology.

C. Clinical Impact – Mortality is often high, depending on the type and grade of the tumor.

D. Treatment Considerations
    1. Corticosteroids for brain edema
        a. Dexamethasone commonly used for vasogenic edema associated with tumors
            i. Reduces peritumoral edema and symptoms associated with increased ICP. Temporarily reduces symptoms (neurologic dysfunction, seizures, headache)
            ii. Dose is commonly 4 mg intravenously every 6 hours.
            iii. May use other corticosteroids at comparable doses
        b. Use of acid-suppressive agents (famotidine) may help with concomitant steroid use to reduce the risk of GI complications (Wien Med Wochenschr 1988;138:97-101).
        c. Consideration for glycemic control
        d. Induction of phenytoin metabolism (because of increased metabolic rate)
        e. Metabolism induced by phenytoin (because of increased cytochrome P450 [CYP] activity)

2. VTE prophylaxis and treatment
    a. High risk of VTE
    b. Consider using combination pharmacologic/mechanical prophylaxis.
    c. Enoxaparin is superior to warfarin for the treatment of VTE in oncology patients.
3. Seizure prophylaxis
    a. Not typically indicated
    b. Around 50% of patients with primary brain tumor present with seizure, which must be treated with anticonvulsant medications.
    c. Phenytoin, carbamazepine, and levetiracetam are often recommended; however, caution should be used with agents that have been associated with Stevens-Johnson syndrome/toxic epidermal necrolysis (e.g., phenytoin, carbamazepine) in patients who also require radiation therapy.
    d. Hepatic CYP enzyme-inducing agents, including phenytoin, should be used with caution because of possible drug interactions with chemotherapy.

XIII. CRITICAL ILLNESS POLYNEUROPATHY

A. Epidemiology
1. Exact incidence is unknown because of inconsistent monitoring and diagnosis.
2. May be as high as 60% in patients with acute respiratory distress syndrome, 77% in long ICU stay (greater than 7 days), 80% in patients with multiorgan failure

B. Diagnosis/Pathogenesis – Diagnostic tests
1. Typically suspected when patients do not wean well from the ventilator or if their limbs are weak/flaccid
2. Electrophysiologic studies or muscle biopsy may provide a more precise diagnosis. Differential diagnosis includes evaluation for critical illness myopathy, Guillain-Barré syndrome, electrolyte abnormalities

C. Causes
1. The cause of critical illness polyneuropathy is unknown, but several hypotheses exist.
    a. Mitochondrial dysfunction in critical illness may cause energy stress in vulnerable neurons.
    b. Microcirculatory ischemia
    c. Protein catabolism in severe critical illness/immobility may cause muscle wasting.
2. Often associated with:
    a. Sepsis
    b. Multiorgan dysfunction
    c. Hyperglycemia
    d. Renal failure
    e. Neuromuscular blockade
    f. Duration of vasopressor or corticosteroid therapy
    g. Duration of ICU stay

D. Clinical Impact
1. Limb and diaphragm weakness may persist for weeks to months.
2. About 33% of patients with critical illness polyneuropathy ultimately cannot independently ambulate or breathe.

E. Treatment Considerations
   1. No specific treatments have been shown effective.
   2. Intravenous immunoglobulin may play a role (Lancet Neurol 2008;7:136-44).
   3. Intensive glycemic control may reduce critical illness neuropathy.
   4. Passive mobilization/early physical therapy in the ICU
   5. Daily awakening/less time on the ventilator
   6. Limiting risk factors as much as possible

XIV. GUILLAIN-BARRÉ SYNDROME

A. Epidemiology
   1. 1.11 cases per 100,000 person-years
   2. Men > women (almost 2:1)

B. Diagnosis/Pathogenesis – Diagnostic tests
   1. Bilateral symmetric progressive weakness of limbs
   2. Generalized hyporeflexia or areflexia
   3. Nerve conduction studies may provide a more precise diagnosis.

C. Causes
   1. Typically associated with *Campylobacter jejuni* infection. Also associated with Epstein-Barr virus, varicella-zoster, and *Mycoplasma pneumoniae* infections
   2. Swine flu vaccine in 1976 caused increased risk of Guillain-Barré syndrome. The same increased risk has not been associated with other seasonal influenza vaccines.

D. Clinical Impact
   1. Progressive weakness over 3–4 weeks
   2. 20% of patients remain severely disabled.
   3. Mortality rate around 5%
   4. Respiratory failure
   5. Autonomic dysfunction resulting in arrhythmia, hypertension, hypotension
   6. Neuropathic pain

E. Treatment Considerations (N Engl J Med 1992;326:1123-9)
   1. Intravenous immunoglobulin versus plasma exchange
   2. Therapies are essentially equivalent.
   3. Plasma exchange: Five treatments over 2 weeks
   4. Intravenous immunoglobulin: 0.4 g/kg intravenously daily × 5 days
   5. Combination of therapies no better than single therapy alone
   6. Benefit of repeated treatments is unclear.
   7. Steroids are not particularly effective.

F. Supportive Care
   1. VTE prophylaxis is imperative.
   2. Careful ventilation strategies to minimize barotrauma and prevent pneumonia

3. Dysphagia is common, so enteral feeding access is necessary in most cases.
4. Neuropathic pain is common; use opiates with caution to avoid respiratory depression.
5. Euvolemia to minimize autonomic instability

XV. MYASTHENIA CRISIS

A. Epidemiology
   1. Annual incidence of myasthenia gravis is 1 or 2 per 100,000.
   2. 15%–20% of patients with myasthenia gravis will develop myasthenia crisis within the first year of illness.

B. Diagnosis/Pathogenesis
   1. Patients with myasthenia crisis typically present with respiratory failure caused by muscle weakness.
   2. Autoimmune disease typically targeting acetylcholine receptors at the neuromuscular junction
   3. Myasthenia crisis is usually preceded by a predisposing factor.
      a. Respiratory infection
      b. Emotional stress
      c. Aspiration
      d. Changes in myasthenia gravis medication regimen
      e. Addition of medications that may oppose acetylcholine effects at the neuromuscular junction and exacerbate myasthenia gravis symptoms (Box 3)
      f. Other physiologic stress (trauma, surgery)

Box 3. Pharmacologic Agents Associated with Myasthenia Crisis

Aminoglycosides	Neuromuscular blockers
β-Blockers	Procainamide
Fluoroquinolones	Quinidine
Macrolides	Tetracyclines
Magnesium	Verapamil

C. Treatment Considerations
   1. Intravenous immunoglobulin 0.4 g/kg/day × 3–5 days or plasmapheresis 20–25 mL/kg plasma × 5 exchanges every other day × 10 days
      a. Similarly effective; can choose according to patient risk factors, etc.
   2. Corticosteroids moderately effective (e.g., prednisone 60–100 mg/d for 2–4 weeks, followed by slow taper)
   3. Supportive Care – Consider discontinuing cholinergic therapies while the patient is acutely ill. May increase pulmonary secretions and complicate ventilator/ICU management

XVI. SEROTONIN SYNDROME (NEUROCRIT CARE 2014;21:108-13)

A. Presentation
   1. Autonomic hyperactivity (hypertension, tachycardia)
   2. Mental status changes

3. Hyperthermia – Diaphoresis
4. Seizures
5. Neuromuscular abnormalities
6. Rigidity
7. Hyperreflexia and clonus
8. Shivering
9. Diarrhea

Table 15. Medications Associated with Serotonin Syndrome

Drug Class	Associated Drugs
Antiemetics	Ondansetron, granisetron, metoclopramide
Antiepileptics	Lamotrigine, carbamazepine, valproic acid
Anti-migraine	Triptans, ergotamine, methylergonovine
Herbal products	St. John's wort, tryptophan, ginseng
Illicit drugs	Methamphetamine, amphetamine, ecstasy (MDMA), psilocybin, LSD
Opioids	Fentanyl, meperidine, methadone, dextromethorphan, tramadol[a]
Psychiatric	
SSRIs	Citalopram,[a] fluoxetine,[a] sertraline,[a] escitalopram, paroxetine
SNRIs	Venlafaxine, duloxetine
TCAs	Amitriptyline, clomipramine, imipramine
MAOIs	Phenelzine, rasagiline, selegiline
Others	Bupropion,[a] buspirone, lithium, trazodone
Miscellaneous	Linezolid, cyclobenzaprine, methylene blue, fluconazole, chlorpheniramine

[a]Top 5 agents implicated in serotonin syndrome

MAOIs = monoamine oxidase inhibitors; SNRIs: serotonin and norepinephrine reuptake inhibitors; SSRIs: selective serotonin reuptake inhibitors; TCAs: tricyclic antidepressants.

Information from: Mikkelsen N, Damkier P, Arnspang Pedersen S. Serotonin syndrome – a focused review. Basic Clin Pharmacol Toxicol 2023;133:124-9; Pedavally S, Fugate JE, Rabinstein AA. Serotonin syndrome in the intensive care unit: clinical presentations and precipitating medications. Neurocrit Care. 2014;21:108-13.

B. Treatment Considerations
   1. Removal of precipitating drugs/factors – Mild symptoms may resolve in 24–72 hours; if caused by antidepressants, may take weeks to resolve
   2. Control of agitation, seizures, and rigidity – Benzodiazepines
   3. Control of autonomic hyperactivity – Hypotension treatment with direct-acting sympathomimetics
   4. Control of hyperthermia
      a. Cooling blanket
      b. Sedation, neuromuscular paralysis, intubation
      c. Avoid succinylcholine.
   5. Serotonin-2A antagonist blocks serotonin receptors implicated with serotonin syndrome.
      a. Cyproheptadine 12–32 mg/24 hours by mouth or feeding tube. A 12-mg loading dose; then 2 mg every 2 hours as symptoms continue
      b. Chlorpromazine 50–100 mg intramuscularly

XVII. NEUROLOGIC MONITORING DEVICES

A. ICP Monitors
   1. Catheters are typically inserted under sterile conditions at the bedside.
   2. Antibiotic prophylaxis is not useful and should not be used routinely (Neurocrit Care 2016;24:61-81).
   3. Antibiotic-coated catheters are available for patients at high risk of infection.
   4. Temporary catheters
      a. Used primarily in the ICU
      b. Ventriculostomy (a.k.a. external ventricular drain) – Diagnostic and therapeutic
         i. Catheter inserted into frontal horn of lateral ventricle
         ii. Transduces ICP; should be calibrated to zero routinely
         iii. Higher infection rate than with intraparenchymal catheter; insertion is more difficult (particularly with brain swelling)
         iv. Permits drainage of CSF and intraventricular hemorrhage
         v. Permits intraventricular drug administration
      c. Intraparenchymal catheter
         i. Wire that sits in brain tissue
         ii. Transduces ICP
         iii. Low infection rate, fewer complications with insertion
         iv. Cannot recalibrate the catheter, experience "zero drift" in ICP readings after prolonged use. May not be entirely accurate for duration of use
      d. Brain tissue oxygen monitor (Licox)
         i. Intraparenchymal catheter
         ii. Optimal location of placement is not well defined (injured vs. non-injured tissue).
         iii. Transduces partial pressure of brain tissue oxygen
            (a) Goal partial pressure of brain tissue oxygen is usually 25–35 mm Hg.
            (b) Concept similar to Svo_2 (mixed venous oxygen saturation) values systemically
         iv. Low values reflect increased ICP or reduced oxygen delivery.
         v. Typically, will be used in combination with other monitoring modalities
      e. Subarachnoid bolt
         i. Single-lumen screw inserted through a burr hole into the subarachnoid space
         ii. Transduces ICP
         iii. Associated with increased CNS infection

B. EEG
   1. Scalp electrodes are placed externally.
   2. Permits evaluation of cortical electrical activity
   3. Standard of care for seizure monitoring or if initiating phenobarbital for ICP control to monitor for attainment of burst suppression
   4. Newer, less cumbersome devices are currently available that may improve accessibility.
      a. Point-of-care EEG systems (e.g., Ceribell) may be useful for faster set up and triage of seizures compared with traditional EEG set ups.

C. Transcranial Doppler
   1. Ultrasound of intracranial vessels
   2. Used in monitoring for cerebral blood flow velocity or vasospasm
   3. Threshold values
      a. 125 cm/second may suggest vasospasm.
      b. 200 cm/second typically suggests severe vasospasm.

c. Lindegaard ratio: Ratio of target blood vessel (usually middle cerebral artery) to carotid (internal carotid artery) blood velocity transcranial Doppler values – value of 3 or more suggests vasospasm

D. Bispectral Index (BIS) Monitor
   1. Scalp electrodes are placed externally.
   2. Uses EEG information to derive a number; should not be substituted for an EEG in patients with seizures
   3. BIS 0–100 (100 being completely wakeful)
   4. Little correlation with BIS values and ICP control or extent of pharmacologic coma

E. Pulpilometer
   1. A handheld device that uses high-speed camera and computing technology to provide quantitative, reproductible, and precise measurements regarding the pupilary light reflex.
   2. The device reports measurements via Neurological Pulp Index derived from a proprietary formula. Values range from 0.0 to 4.9, and a value of less than 3.0 is considered an abnormal or sluggish response.

XVIII. IMPACT OF SOCIAL DETERMINANTS OF HEALTH (SDOH)

A. SDOH are conditions in the places where people live, learn, work, and play that affect health risks and outcomes.

B. Disparates noted in select neurological injuries:
   1. Stroke incidence: Low education, low income, living in an impoverished area, residence in an area with few healthcare services, Southeastern residence, living in rural areas, social isolation, and lacking health insurance
   2. Traumatic brain injury (TBI):
      a. American Indian and Alaska Native have higher rates of TBI-related hospitalizations and deaths compared with other racial or ethnic groups. Factors contributing to this disparity include motor vehicle accidents, substance abuse, and difficulties accessing healthcare.
      b. Military service members saw an increase in TBI because of conflicts from 2005 to 2018
      c. Research suggests almost half of people in correctional or detention facilities have a history of TBI and may face challenges with accessing TBI-related care.
      d. Those who experience homelessness are up to 10 times more likely to have a history of moderate to severe TBI. They have worse overall physical and mental health.
      e. Survivors of intimate partner violence who have experienced a TBI are more likely to be diagnosed with post traumatic stress disorder, insomnia, and depression.
      f. Individuals living in rural areas have a greater risk of death following TBI compared with those living in urban areas because of travel time to receive medical care and less access to Level I Trauma Center.

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ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: C**
Answer C is correct because midazolam (together with lorazepam) is recommended by the status epilepticus guidelines. Answers A, B, and D are incorrect because phenytoin is less effective than lorazepam as the initial agent. Although valproic acid and levetiracetam have not been formally compared with lorazepam as the initial agent for status epilepticus, their use is supported by less clinically rigorous evidence.

2. **Answer: B**
Answer B is correct because clinical evidence supports the safety and efficacy of osmotherapy as a first-line therapy in this situation. Hypertonic saline would be more appropriate than mannitol because of the patient's relatively low serum sodium concentration and elevated SCr (mannitol is cleared renally and is thus not optimal for patients with renal dysfunction, making Answer A incorrect). Answers C and D (pentobarbital and midazolam) are not ideal for this patient because of the likelihood of hypotension.

3. **Answer: B**
Answer B is correct because warfarin is well reversed by 4F-PCC products in a much more timely and complete manner than vitamin K in the acute setting. Answer A is incorrect; although blood pressure control is important for this patient, amlodipine is unlikely to have timely effects immediately after ICH. Answer C is incorrect because platelets are minimally effective for reversing ibuprofen. Answer D is incorrect; rFVIIa is not recommended for reversal of warfarin because of thrombosis risks.

4. **Answer: B**
Answer B is correct because nicardipine is recommended for reducing blood pressure after ICH, and the threshold for treatment is correct according to the INTERACT-2 and ATACH-2 studies. Answer A is incorrect; although clevidipine may be considered in this case, the typical goal blood pressure target after ICH is a SBP 140-150 mm Hg. Answers C and D are incorrect; although labetalol and esmolol also reduce blood pressure, the optimal SBP goal after ICH 140-150 mm Hg.

5. **Answer: A**
Answer A is correct because nimodipine is the only agent with an FDA indication for preventing ischemic complications related to SAH. Answer B is incorrect because prophylactic Triple-H therapy or variants thereof do not prevent ischemic complications; rather, hyperperfusion therapies are used when vasospasm develops. Answer C is incorrect because in clinical trials, the efficacy of statins for preventing vasospasm has failed. Answer D is incorrect because aminocaproic acid may in fact increase the risk of stroke in patients with SAH.

6. **Answer: A**
Answer A is correct because induction of hypertension with a vasopressor such as norepinephrine appears to improve cerebral perfusion. Titrating the infusion to MAP values that result in improved neurologic symptoms is often necessary. Answer B is incorrect because data analyses are limited to support transfusing blood to a high hemoglobin (in fact, blood transfusion appears to be a risk factor for vasospasm). In addition, fluid resuscitation to hypervolemic levels is not beneficial. Answer C is incorrect because when hypervolemia is compared with euvolemia, neurologic outcomes are no different, but patients receiving hypervolemia develop more pulmonary edema. Answer D is incorrect because milrinone is not first-line therapy for vasospasm.

7. **Answer: B**
Answer B is correct because the anti-Xa activity concentration is the laboratory value that best correlates with rivaroxaban activity. Answers A and C are incorrect because neither INR nor activated partial thromboplastin time is typically affected by rivaroxaban alone. Answer D is incorrect because the VerifyNow PRU test measurement is more specific to antiplatelet agents such as aspirin or clopidogrel.

8. **Answer: A**
Answer A is correct because the most consistent reversal effects, although with low-quality evidence, occur with 4F-PCCs. Answer B is incorrect because the appropriate bolus dose of andexanet would be 400 mg followed by a 2-hour continuous infusion, given the dose and time of the last dose. In addition, available data are currently limited regarding the periprocedural use of andexanet. Answer C is incorrect because fresh frozen plasma does not reverse factor Xa inhibitors reliably. Answer D is incorrect; although factor VII may have some usefulness, reversal is incomplete, and factor VII is associated with an increased risk of thrombosis.

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: B**

Answer B is correct because sodium supplementation is effective for treating hypovolemic and euvolemic hyponatremia. Although inducing hypervolemia is no longer advocated, ensuring euvolemia is important. Answers A and C are incorrect because water restriction or tolvaptan is undesirable in a patient who is 4 days after ictus for SAH because of the importance of maintaining adequate cerebral perfusion. Hyponatremia may be harmful in a patient with a new stroke such as SAH caused by cerebral edema, making Answer D incorrect.

2. **Answer: A**

Answer A is correct because vancomycin effectively covers MRSE, and the intraventricular dose of 10 mg daily is appropriate. Answer B is incorrect because gentamicin is less likely to be effective alone for MRSE and is more associated with seizures than other aminoglycosides. Answer C is incorrect because ampicillin (or other penicillins) should not be given by the intraventricular route, given the risk of seizures. Answer D is incorrect because this patient has a treatment-refractory, device-related CNS infection. In this instance, intraventricular antimicrobials may be considered.

3. **Answer: B**

Answer B is correct; although the optimal CPP varies with each individual, the typical recommended target range is 60–70 mm Hg. Patients with a TBI have gastric intolerance but benefit greatly from early enteral nutrition, making Answer A incorrect. Answer C is incorrect because dextrose-containing fluids may increase cerebral edema in TBI and lower the serum sodium concentration. Answer D is incorrect because high-dose methylprednisolone therapy increases mortality in patients with a TBI.

4. **Answer: B**

Answer B is correct because this patient meets the criteria for receiving alteplase and has no obvious contraindications. The more timely the alteplase administration, the more likely the patient will benefit (and the less risk). Answer A is incorrect because aspirin should be initiated within the first 24–48 hours after stroke, but not necessarily immediately. Answer C is incorrect because the blood pressure may be slightly elevated (SBP less than 185 mm Hg, DBP less than 110 mm Hg) before alteplase administration or immediately after stroke in general (so-called permissive hypertension to ensure adequate cerebral perfusion). Nicardipine is not necessary for this patient at this point. Answer D is incorrect because reversal of warfarin with vitamin K is not recommended for an acute thrombosis in the brain.

5. **Answer: C**

Answer C is correct; the current guidelines do not recommend high-dose methylprednisolone therapy because of the inconsistency in beneficial effects and the relatively consistent risk of adverse effects (GI bleeding, infection) shown in clinical trials. Answer B is incorrect because high-dose methylprednisolone does not augment spinal perfusion. Answers A and D are incorrect; although the NASCIS-III study showed some potential benefit for patients who received a bolus and a 47-hour infusion when high-dose methylprednisolone therapy was initiated 3–8 hours after injury, particularly for incomplete injuries, the therapy is no longer recommended.

6. **Answer: A**

Answer A is correct; of the agents listed, buspirone is the only one that increases CNS serotonin concentrations. Answers B–D would not be expected to increase CNS serotonin concentrations. Cyproheptadine is a potential therapeutic agent for patients with serotonin syndrome.

7. **Answer: B**

Answer B is correct; given this patient's change in neurologic status, vital signs, and monitoring values, she is likely having a cerebral vasospasm with increased ICP. Therapy should be targeted at optimizing cerebral perfusion (fluid bolus or increase in MAP) and reducing ICP. Hypertonic saline will address both issues, causing an increase in intravascular volume to improve perfusion and possibly an increase in blood pressure while lowering ICP through osmotic effects. Answer A is incorrect because verapamil must be given superselectively in the angiography suite. One possible adverse effect of verapamil is cerebral vasodilation, which could lead to increased ICP, and ICP would be undesirable in this patient right now. Answer C is incorrect; although mannitol may decrease ICP, it also causes diuresis,

which is undesirable in a patient with ongoing cerebral vasospasm. Answer D is incorrect; although a blood transfusion might theoretically increase oxygen-carrying capacity to the brain (enhancing perfusion), a blood transfusion may in fact cause a risk of cerebral vasospasm. In addition, it is unlikely to affect ICP.

8. **Answer: D**

Answer D is correct because propofol is an almost-ideal agent for sedation with its quick onset and offset of activity and relative lack of dependence on organ function for clearance; however, boluses and large dose titrations should be avoided because of the risk of hypotension. Answers A and C are incorrect; long-acting sedating agents such as lorazepam and morphine are undesirable in patients with a neurologic injury because of their propensity to obscure the neurologic examination for prolonged periods. Answer B is incorrect; midazolam (and other benzodiazepines) is associated with increased delirium. In addition, midazolam may accumulate when used for a prolonged period because of its long context-sensitive half-life.